

Modular Forms

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1. Elementary Modular Forms

1.1. Basic Definitions and Notation

Definition 1.1.1 (*Modular Forms*) A function

$$f : \mathbb{H} \rightarrow \mathbb{C} \quad (1)$$

is called **modular** (of level 1) of weight $k \in \mathbb{Z}$ if for

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) \quad (2)$$

we have

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z) \quad (3)$$

Other terminology: Automorphic Forms.

Remark 1.1.2

1. If $z \in \mathbb{H}$ and $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{R})$ then $\frac{az+b}{cz+d} \in \mathbb{H}$.

$$\Im\left(\frac{az+b}{cz+d}\right) = \frac{\Im(z)}{|cz+d|^2} > 0 \quad (4)$$

by definition.

2. The set of weight k modular forms is a $\mathbb{C}$ -vector space, and if f_1 has weight k_1 and if f_2 has weight k_2 then $f_1 f_2$ has weight $k_1 k_2$. $\Rightarrow$ Construction of weight 0 functions as ratios of f_i of same weight, linearly independent.

Definition 1.1.3 (*Action of $SL_2(\mathbb{R})$ on $\mathbb{H}$ and modular forms*) $\forall g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{R}), \forall z \in \mathbb{H}$

$$g \cdot z = \frac{az+b}{cz+d} \quad (5)$$

$$\forall f : \mathbb{H} \rightarrow \mathbb{C}$$

$$(f |_k g)(z) = f\left(\frac{az+b}{cz+d}\right) \frac{1}{(cz+d)^k} \quad (6)$$

Note the action on the z is a left action and the action on the functions is a right action. So f is **modular** of weight $k \Leftrightarrow$ if f is invariant under the weight k right action of $SL_2(\mathbb{Z})$.

Lemma 1.1.4 (*Transitivity of action of $SL_2(\mathbb{R})$*) $SL_2(\mathbb{R})$ acts transitively on $\mathbb{H}$. In fact $B = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \mid a \in \mathbb{R}^*, b \in \mathbb{R} \right\}$ acts transitively. Moreover $SL_2(\mathbb{R})^i = SO_2(\mathbb{R})$

Proof

1. We will show that $\forall z = x + iy \in \mathbb{H}, \exists M \in B \ni M \cdot z = i$. We want to find $(a, b) \in \mathbb{R}^\times \times \mathbb{R} \ni i = \frac{a(x+iy)+b}{\frac{1}{a}} = a(ax+b) + a^2iy$. Thus $a = \sqrt{\frac{1}{y}}$ and $b = -x\sqrt{\frac{1}{y}}$. Thus

$$M = \begin{pmatrix} \sqrt{\frac{1}{y}} & -x\sqrt{\frac{1}{y}} \\ 0 & \sqrt{y} \end{pmatrix} \quad (7)$$

and $M \cdot z = i$. Since it is a group action, the action is transitive.

2. We need to show that $M \cdot i = i \Rightarrow M \in SO_2(\mathbb{R})$. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$\frac{ai + b}{ci + d} = i \Rightarrow ai + b = di - c \quad (8)$$

Thus $a = d$ and $c = -b$. Thus $M = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$.

$$\begin{aligned} MM^T &= \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \\ &= \begin{pmatrix} a^2 + b^2 & 0 \\ 0 & a^2 + b^2 \end{pmatrix} \end{aligned} \quad (9)$$

Furthermore note $M^T \in SL_2(\mathbb{R}) \Rightarrow MM^T \in SL_2(\mathbb{R}) \Rightarrow (a^2 + b^2)^2 = 1 \Rightarrow a^2 + b^2 = 1 \Rightarrow MM^T = I$. Thus $M \in SO_2(\mathbb{R})$. Thus $SL_2(\mathbb{R})^i \subset SO(\mathbb{R})$. Let $\begin{pmatrix} a & b \\ -b & a \end{pmatrix} \in SO(\mathbb{R})$.

$$\begin{aligned} \frac{b + ai}{a - bi} &= \frac{(b + ai)(a + bi)}{a^2 + b^2} \\ &= \frac{ab - ab + i(a^2 + b^2)}{a^2 + b^2} \\ &= i \end{aligned} \quad (10)$$

Thus $SO(\mathbb{R}) \subset SL_2(\mathbb{R})^i$. Thus $SL_2(\mathbb{R})^i = SO(\mathbb{R})$. $\square$

Lemma 1.1.5 (NAK Decomposition of $SL_2(\mathbb{R})$)

Let $N = \left\{ \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \right\}$, $A = \left\{ \begin{pmatrix} a & \\ & a^{-1} \end{pmatrix} \right\}$, and $K = SO_2(\mathbb{R})$.

$$SL_2(\mathbb{R}) = NAK \quad (11)$$

Proof Let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{R})$. Let $z = x + iy = g \cdot i$. We want to show $\exists(N, A) \in N \times A \ni NA \cdot i = z$. Let $N = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ and $A = \begin{pmatrix} \sqrt{y} & \\ & \frac{1}{\sqrt{y}} \end{pmatrix}$. Thus $NA \cdot i = z$. Let $K = (NA)^{-1}g$. Thus $g = NAK$. Thus $SL_2(\mathbb{R}) = NAK$. $\square$

So that is how $SL_2(\mathbb{R})$ acts on $\mathbb{H}$, but what we need to understand is how $SL_2(\mathbb{R})$ acts on $\mathbb{H}$.

Analogy: $\mathbb{R}$ transitively acts on $\mathbb{R}$ by translation. $\mathbb{Z} \subset \mathbb{R}$ acts on $\mathbb{R}$ by translation but the action is no longer transitive, each orbit is characterized by the fractional part of the number thus for any function invariant under translation by $\mathbb{Z}$ we only need to understand the behavior on $[0, 1)$. Thus $[0, 1)$ is the fundamental domain of $\mathbb{Z}$. Similarly $SL_2(\mathbb{R})$ acts transitively on $\mathbb{H}$, but $SL_2(\mathbb{Z})$ acts in the same way but isn't transitive. However its orbits can also be completely characterized, so what is the fundamental domain of $SL_2(\mathbb{Z})$?

Theorem 1.1.6 (*Fundamental Domain of $SL_2(\mathbb{Z})$*) Let

$$D_F := \left\{ z \in \mathbb{H} \mid |\Re(z)| < \frac{1}{2} \text{ and } |z| \geq 1 \right\} \quad (12)$$

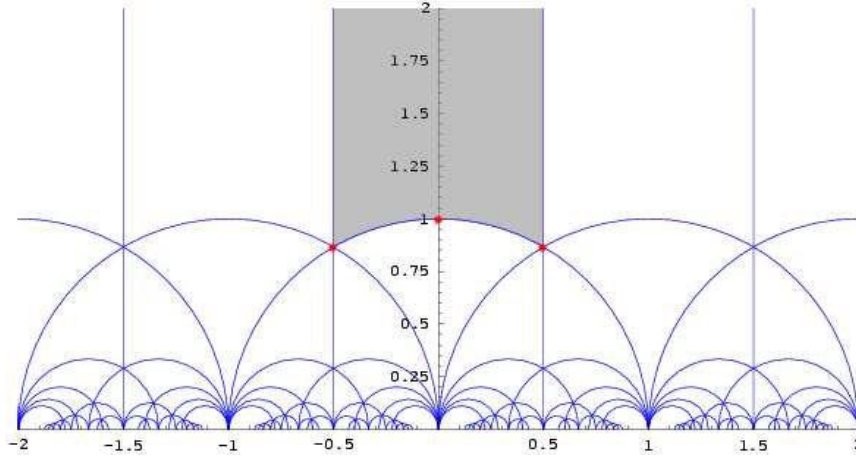


Figure 1: The fundamental domain of $SL_2(\mathbb{Z})$

1. Every $SL_2(\mathbb{Z})$ -orbit intersects $D (\Leftrightarrow \forall z \in \mathbb{H}, \exists g \in SL_2(\mathbb{Z}) \ni g \cdot z \in D_F)$
2. If $z_1, z_2 \in D_F^o = \{z \in \mathbb{H} \mid |\Re(z)| < \frac{1}{2} \text{ and } |z| > 1\}$ the orbits of z_1 or z_2 are equal or disjoint
3. If $z_1, z_2 \in D$ and $gz_1 = z_2$ for $g \in SL_2(\mathbb{Z})$ then:
 - $\Re(z_1) = -\frac{1}{2}$ and $g = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$
 - $\Re(z_1) = \frac{1}{2}$ and $g = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$
 - $|a| = 1$ and $g = \pm \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$
4. For $z \in D$,

$$\text{Stab}_{SL_2(\mathbb{R})}(z) = \begin{cases} \{\pm 1\} & \text{if } z \notin \left\{i, e^{\frac{\pi i}{3}}, e^{\frac{2\pi i}{3}}\right\} \\ \pm \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} & \text{if } z = i \\ \pm \left\langle \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \right\rangle & \text{if } z = e^{2\pi i/3} \\ \pm \left\langle \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \right\rangle & \text{if } z = e^{\pi i/3} \end{cases} \quad (13)$$

Proof Proof of (1), and (2). Key point: $SL_2(\mathbb{Z})$ is discrete in $SL_2(\mathbb{R})$.

1. We will prove something seemingly stronger, that any Γ -orbit intersects D_F where $\Gamma = \left\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right\rangle$. Let $z \in \mathbb{H}$. **WTS: $\exists g \in \Gamma \ni \mathfrak{I}(gz)$ is maximal.**

$$\mathfrak{I}(g \cdot z) = \frac{\mathfrak{I}(z)}{|cz + d|^2} \quad (14)$$

If we want $\mathfrak{I}(gz)$ to be maximal, we want $|cz + d|^2 = (c(x + iy) + d)(c(x - iy) + d) = (cx + d)^2 + c^2y^2$ to be minimal. Thus look at the set, $\{(c, d) \in \mathbb{Z}^2 \mid (c, d) \neq (0, 0), |cz + d| \leq 1\}$ is a finite set. Then take an element with $|cz + d|$ minimal in the finite set. The set and its intersection with Γ will not be empty as identity will be in both.

Let $g \in \Gamma$, be the element with this property such that $g \cdot z$ is maximal. WLOG, $|z| \geq 1$, for $|z| < 1$, take $z = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} z = -\frac{1}{z}$ since $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in \Gamma$. Then $\exists k \in \mathbb{Z} \ni |\Re(z) - k| \leq \frac{1}{2}$ and then replacing g with $g' = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} g$ gives an element in $\Gamma \ni g'z \in D_F$.

2. Let $z \in D_F$ and $g \in \text{SL}_2(\mathbb{Z}) \ni g \cdot z \in D_F$.

1. $\mathfrak{I}(gz) \geq \mathfrak{I}(z)$: Then $|cz + d|^2 = (cx + d)^2 + c^2y^2 \leq 1$. But $y \geq \frac{\sqrt{3}}{2}$ with equality iff $z = e^{\pm \frac{2\pi i}{3}}$. Thus $1 \geq c^2y^2 \geq \frac{3}{4}c^2 \Rightarrow c^2 \leq \frac{4}{3} \Rightarrow c^2 \leq 1$ since $c \in \mathbb{Z}$. If $c = 0 \Rightarrow g \cdot z$ is a integer translate of z thus $\Re(z) = \frac{1}{2}$ or $\frac{1}{2}$ and $g = \pm \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ or $\pm \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$. Or $c = 1$, then $|z + d| \leq 1$, you get the further cases in (3).

2. $\mathfrak{I}(gz) < \mathfrak{I}(z)$: replace (g, z) by (g^{-1}, gz) and use 1.

□

Corollary 1.1.7 $\Gamma = \text{SL}_2(\mathbb{Z})$, ie $\forall g \in \text{SL}_2(\mathbb{Z})$ g is a product of $\pm 1, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Proof Let $g \in \text{SL}_2(\mathbb{Z})$. Let $z = g(2i)$. By Theorem 1.1.6.1, $\exists h \in \Gamma \ni z' = h \cdot z \in D_F$. Then $z', 2i$ are in D are $\text{SL}_2(\mathbb{Z})$ -equivalent. By part 2,3, $z' = 2i$ and $g = \pm h \in \Gamma$. □

Remark 1.1.8

1. $\{g \in \text{SL}_2(\mathbb{R}) \mid \forall z, gz = z\} = \{\pm 1\}$ one can always replace $\text{SL}_2(\mathbb{R})$ and $\text{SL}_2(\mathbb{Z})$ with $\text{PSL}_2(\mathbb{R})$ and $\text{PSL}_2(\mathbb{Z})$.
2. D_F is not compact. This creates a lot of analytic complications; on the other hand this gives a distinguished infinity direction, namely “up”.

1.2. First Examples of Modular Forms

We will construct a modular form of a given weight k (if possible), which are meromorphic or holomorphic.

Proposition 1.2.1 For a group G acting on X , if you have $f : X \rightarrow \mathbb{C}$ that is invariant under the action of $H < G$. Then the function $F : X \rightarrow \mathbb{C}$ defined by

$$F(x) = \sum_{g \in H \backslash G} f(gx) \quad (15)$$

if absolutely convergent is (well defined) and G invariant function.

Proof First note $\forall g \in H \backslash G$, $f(gx)$ is well defined because replacing g with hg gives

$$f(hgx) = f(h(gx)) = f(gx) \quad (16)$$

by the assumption that f is H -invariant.

Then for $\gamma \in G$,

$$\begin{aligned} F(\gamma x) &= \sum_{g \in H \backslash G} f(g\gamma x) \\ &= \sum_{g' \in H \backslash G} f(g'x) = F(x) \end{aligned} \quad (17)$$

□

Example 1.2.2 $G = \mathbb{Z}$, $X = \mathbb{R}$, $H = \{0\}$. $f : \mathbb{R} \rightarrow \mathbb{C}$. Then $F(x) = \sum_{n \in \mathbb{Z}} f(n+x)$ is 1-periodic.

We will apply this to

$$\begin{aligned} G &= \mathrm{SL}_2(\mathbb{Z}) \\ H = N &= \left\{ \pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \mid n \in \mathbb{Z} \right\} \end{aligned} \quad (18)$$

(Note $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} z = n + z$). Let

$$e(z) = e^{2\pi i z} \quad (19)$$

, e is invariant under integer translation and is holomorphic on $\mathbb{H}$.

Definition 1.2.3 (Translation Subgroup) Let $T = \left\{ \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \mid n \in \mathbb{Z} \right\} < \mathrm{SL}_2(\mathbb{Z})$ be the translation subgroup. Let $\bar{T} = \left\{ \pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \mid n \in \mathbb{Z} \right\}$.

Definition 1.2.4 (Poincaré Series)

$$P_{m,k}(z) = \sum_{g \in \bar{T} \backslash G} (cz + d)^{-k} e(m(g \cdot z)) \quad (20)$$

is a **Poincaré Series**

Definition 1.2.5 (j Helper Function) For $k \in \mathbb{Z}$, let $j_k : \mathrm{GL}_2(\mathbb{R}) \times \mathbb{C} \rightarrow \mathbb{C}$ where

$$j_k \left(\begin{pmatrix} * & * \\ c & d \end{pmatrix}, z \right) := (cz + d)^{-k} \quad (21)$$

.

Thus

$$(f|_k g)(z) = j_k(g, z) f(g \cdot z) \quad (22)$$

Proposition 1.2.6 If $P_{m,k}$ converges absolutely then $P_{m,k}$ is a weight k modular form

Proof

$$(cz + d)^{-k} e(m(g \cdot z)) = (e_m |_k g)(z) \quad (23)$$

where $e_m(z) = e(mz)$.

$\forall \gamma \in \mathrm{SL}_2(\mathbb{Z})$,

$$\begin{aligned} (P_{m,k} |_k \gamma)(z) &= j_{k(\gamma,z)} P_{m,k}(\gamma z) \\ &= j_{k(\gamma,z)} \sum_{g \in H \backslash G} (e_m |_k g)(\gamma z) \\ &= \sum_{g \in H \backslash G} j_{k(\gamma,z)} (e_m |_k g)(\gamma z) \\ &= \sum_{g \in H \backslash G} (e_m |_k g |_k \gamma)(z) \\ &= \sum_{g \in H \backslash G} (e_m |_k g\gamma)(z) \\ &= \sum_{g' \in H \backslash G} (e_m |_k g')(z) = P_{m,k}(z) \end{aligned} \quad (24)$$

□

Lemma 1.2.7 Let $\alpha : \mathrm{SL}_2(\mathbb{Z}) \rightarrow \mathbb{Z}^2$ where $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \rightarrow (c, d)$. α factors through $T \backslash \mathrm{SL}_2(\mathbb{Z})$

Proof $\forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ and $\forall n \in \mathbb{Z}$

$$\begin{aligned} \alpha\left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) &= \alpha\left(\begin{pmatrix} * & * \\ c & d \end{pmatrix}\right) \\ &= (c, d) \end{aligned} \quad (25)$$

Thus α is invariant under right multiplication by elements of T .

Suppose

$$\alpha\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = \alpha\left(\begin{pmatrix} a' & b' \\ c & d \end{pmatrix}\right) \quad (26)$$

Then $1 = ad - bc = a'd - b'c \Rightarrow d(a - a') = c(b - b')$. Since $(c, d) = 1$, we get $d \mid (b - b')$ and $c \mid (a - a')$. Thus $b = b' + kd$ and $a = a' + lc$. Thus

$$d(a - a') = dlc = ckd = c(b - b') \quad (27)$$

Thus $l = k$. Thus

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a' & b' \\ c & d \end{pmatrix} \quad (28)$$

Thus α factors through $T \backslash \mathrm{SL}_2(\mathbb{Z})$

□

Lemma 1.2.8 For $z \in \mathbb{H}$ and $k > 2$, the series

$$\sum_{(c,d) \neq (0,0)} |cz + d|^{-k} \quad (29)$$

converges absolutely and locally uniformly.

Proof Note that $(c, d) \rightarrow \max(|c|, |d|)$ and $(c, d) \rightarrow |cz + d|$ are both norms on $\mathbb{R}^2$. The second because since $z \in \mathbb{H}$, $cz + d = 0 \Leftrightarrow (c, d) = (0, 0)$. By the equivalence of norms on $\mathbb{R}^2$. Thus

$$\begin{aligned} |cz + d| &\leq \alpha(z) \max(|c|, |d|) \\ \max(|c|, |d|) &\leq \beta(z) |cz + d| \end{aligned} \quad (30)$$

For two norms in $\mathbb{R}^n$, $|\cdot|_1$ and the second norm be $|\cdot|_2$, we have $|v|_1 \leq c|v|_2$ for $c = \sup_{|v|_1=1} (|v|_2) = \max_{|v|_1=1} (|v|_2)$. So

$$\begin{aligned} \alpha(z) &= \max_{\max(|c|, |d|)=1} (|cz + d|) \\ \beta(z) &= \max_{|cz+d|=1} (\max(|c|, |d|)) \end{aligned} \quad (31)$$

If z varies in a compact subset of $\mathbb{H}$, by property continuous functions α and β are bounded.

Let $r(n) := |\{(c, d) \in \mathbb{Z}^2 \mid n \leq |cz + d| < n + 1\}|$. Define two functions

$$\begin{aligned} A(z) &= \frac{r(0)}{\min_{(c,d) \in \mathbb{Z}^2 - (0,0)} (|cz + d|^k)} \\ B(z) &= \sum_{n \geq 1} \frac{r(n)}{n^k} \end{aligned} \quad (32)$$

Thus,

$$\sum_{(c,d) \neq (0,0)} |cz + d|^{-k} \leq A(z) + B(z) \quad (33)$$

1. For $r(0)$: Since $|cz + d| < 1 \Rightarrow \max(|c|, |d|) \leq \beta(z)$. For any choice of z , there is a fixed finite number of $(c, d) \in \mathbb{Z}^2$ such that $\max(|c|, |d|) \leq \beta(z)$. Thus $r(0) = O(1)$.
2. For $n \geq 1$ $r(n)$: $\max(|c|, |d|) \leq \beta(z)(n + 1)$ thus we have upper bound for $|c|$ and $|d|$. We also have, $\frac{n}{\alpha(z)} \leq \max(|c|, |d|)$. Thus we have roughly speaking $r(n)$ counts the number of (c, d) in the annulus $n \leq |cz + d| \leq n + 1$ which grows proportionally to n . Thus $r(n) = O(n)$.

$\min_{(c,d) \in \mathbb{Z}^2 - (0,0)} (|cz + d|^k)$ bounded below uniformly locally, because $\mathbb{Z} + \mathbb{Z}z$ is discrete in $\mathbb{R}^2$. Thus the series is dominated by

$$c \sum_{n \geq 1} \frac{n}{n^k} = c \sum_{n \geq 1} n^{1-k} \quad (34)$$

Since $k > 2$, the series is locally absolutely convergent. $\square$

Theorem 1.2.9 For $m \geq 0$ and $k \geq 3$, the series $P_{m,k}$ converges absolutely and locally uniformly on $\mathbb{H}$

Proof Note that $\forall z = x + iy \in \mathbb{H}$, $|e(mz)| = e^{-2\pi my} \leq 1$, so the statement will hold for all $m > 0$ if it holds for $m = 0$. Thus it suffices to prove that $P_{0,k}$ converges absolutely and locally uniformly on $\mathbb{H}$.

$$\begin{aligned} \sum_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} |j_k(g, z)| &< \sum_{g \in T \backslash \mathrm{SL}_2(\mathbb{Z})} |j_k(g, z)| \\ &< \sum_{(c,d) \in \mathbb{Z}^2 - (0,0)} |cz + d|^{-k} \text{ (By Injection Lemma)} \end{aligned} \quad (35)$$

The last sum is absolutely and locally uniformly convergent. Thus $\sum_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} |j_k(g, z)|$ is absolutely and locally uniformly convergent. Thus $P_{0,k}$ is absolutely and locally uniformly convergent. $\square$

Remark 1.2.10 For any modular form f since $-I \in \mathrm{SL}_2(\mathbb{Z})$, and $f(-Iz) = f(z) = (-1)^k f(z)$. That means that f is either zero or k is even.

1.3. Fourier Expansions

Idea: Study modular forms “at ∞ ” ie for $\Im(z) \rightarrow \infty$

Theorem 1.3.1 (Existence of Fourier Expansion) Let

$$f : \mathbb{H} \rightarrow \mathbb{C} \quad (36)$$

be a meromorphic modular form of weight k there exists a meromorphic $\tilde{f} : \mathbb{D}^* \rightarrow \mathbb{C}$ such that $f(z) = \tilde{f}(e(z))$

Proof The map $e : D \rightarrow \mathbb{D}^*$ where $\tilde{D} = \{z \in \mathbb{H} \mid |\Re(z)| < \frac{1}{2}\}$ is a conformal equivalence. Thus e^{-1} is also holomorphic. Thus define $\tilde{f} = f \circ e^{-1}$ is a meromorphic on $\mathbb{D}$. Thus $f = \tilde{f} \circ e$. Since $f(z+1) = f|_k \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} (z) = f(z) \Rightarrow$ this holds everywhere on $\mathbb{H}$ using periodicity and continuity. $\square$

Definition 1.3.2 f is a meromorphic modular form of weight k is called **meromorphic or holomorphic** at ∞ if $\tilde{f}$ extends to a meromorphic or holomorphic function $\tilde{f} : \mathbb{D} \rightarrow \mathbb{C}$.

Definition 1.3.3 f is a cusp form of weight k if its a modular form of weight k and $\tilde{f}(0) = 0$.

Definition 1.3.4 M_k is the space of modular forms of weight k which are holomorphic on $\mathbb{H}$ and at ∞ . $S_k = M_k^o$ is the space of cusp forms.

Definition 1.3.5 For $f \in M_k$, $\tilde{f}$ is holomorphic and thus has a Taylor Expansion

$$\tilde{f}(z) = \sum_{n \geq 0} a_n z^n \quad (37)$$

thus

$$f(z) = \sum_{n \geq 0} a_n e(nz) \quad (38)$$

a_n is the **Fourier coefficients at ∞ of f** By uniqueness of Taylor expansion, the sequence (a_n) characterizes the modular forms.

Suppose (a_n) is some arbitrary series which doesn't grow too fast, how can you determine if its the fourier coefficients of some modular form?

It just has to satisfy the extra symmetry of $SL_2(\mathbb{Z})$. Suppose (a_n) is a series with polynomial growth. Then $f(z) = \sum_{n \geq 0} a_n e(nz)$ defines a function $f : \mathbb{H} \rightarrow \mathbb{C}$ holomorphic such that $f(z+1) = f(z)$ (this comes from exponentials).

Since $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ generate $SL_2(\mathbb{Z})$ we just need to check that

$$f\left(-\frac{1}{z}\right) = z^k f(z) \tag{39}$$

for some k to know if f is modular of a given weight k !

1.4. Dimension of M_k and S_k

$\forall z \in \mathbb{H} \cup \{0\}$, let $v_z : \text{hom}(\mathbb{H}, \mathbb{C}) \rightarrow \mathbb{N} \cup \{0\}$ where $f \rightarrow$ order of f at z (if z is a zero $>$ order and < 0 if pole) and for $v_\infty(f) = v_0(\tilde{f})$ for f weight k meromorphic. Thus if $f \in S_k \Rightarrow v_\infty(f) \geq 1$.

Note that $\forall f \in M_k$ and $\forall z \in \mathbb{H}$, $\forall g \in \Gamma$ $f(gz) = j_{-k}(g, z)f(z)$ (Definition 1.2.5 - J Helper Function). $j_{-k}(g, z)$ does not vanish on $\mathbb{H}$. Thus $v_{gz}(f) = v_z(f)$. So $v_z(f)$ can be defined on $\Gamma \setminus \mathbb{H}$.

Theorem 1.4.1 For $f \in M_k$, $f \neq 0$ then the “number of zeroes of f in $\Gamma \setminus \mathbb{H}$ ” with multiplicity is $\frac{k}{12}$. More precisely,

$$v_\infty(f) + \sum_{z \in \Gamma \setminus \mathbb{H} - \{i, \omega, \bar{\omega}\}} v_z(f) + \frac{1}{2}v_i(f) + \frac{1}{2}v_\omega(f) = \frac{k}{12} \quad (40)$$

where $\omega = e^{2\pi i/3}$. $\frac{1}{2}$ and $\frac{1}{3}$ come from the order of the stablizer of i and ω in $\text{PSL}_2(\mathbb{Z})$

Proof Note $\Gamma \setminus H$ is not D_F , but rather you have identified the two verticals and each half of the arc of the unit circle together. So $z \sim w$ if

1. $|\Re(z)| = |\Re(w)| = \frac{1}{2}$
2. $|z| = |w| = 1$ and $\Re(z) = -\Re(w)$
3. **There exists $y_0 > 0 \ni f(z) \neq 0, \forall z \in \mathbb{H} \ni \Im(z) > y_0$:** f is holomorphic at ∞ , so $\tilde{f} : D^* \rightarrow \mathbb{C}$ is holomorphic at 0 and $\tilde{f} \neq 0$. There are two cases:
 1. $\tilde{f}(0) \neq 0$: Since $\tilde{f}$ is holomorphic, $\exists r \in (0, 1) \ni 0 \notin f(\{z \in \mathbb{D} \mid 0 < |z| \leq r\})$.
 2. $\tilde{f}(0) = 0$: Since the zeros of holomorphic functions are isolated, $\exists r \in (0, 1) \ni 0 \notin f(\{z \in \mathbb{D} \mid 0 < |z| \leq r\})$.

Now that means that $f(z) \neq 0$ for $\Im(z) \geq \frac{1}{2\pi} \log(\frac{1}{r})$. Take $y_0 = \max(2\pi \log(\frac{1}{r}), 2)$.

$\forall \varepsilon > 0$, consider the contour Γ_ε below, if f doesn't have 0s on the boundary. Otherwise, add ε dimples in the contour. For $\varepsilon > 0$ small enough, we have

$$\frac{1}{2\pi} \int_{\Gamma_\varepsilon} \frac{f'}{f} dz = \sum_{z \in \text{SL}_2(\mathbb{Z}) \setminus \mathbb{H} - \{i, \omega\}} v_z(f) \text{ (Logarithmic Derivative)} \quad (41)$$

To calculate the integral, we will tranform the integral using $z \rightarrow e(z)$. Thus the top part of the integral

$$\int_{\frac{1}{2} + iy_0}^{-\frac{1}{2} + y_0} \frac{f'}{f} dz = - \int_{\pi}^{-\pi} \frac{f'(e(z))}{f(e(z))} dz = - \int_{\pi}^{-\pi} \frac{\tilde{f}'}{\tilde{f}} dz = -2\pi v_\infty(f) \quad (42)$$

The vertical chunks of the integral cancel each other out.

$$\begin{aligned}
\int_{C_1} \frac{f'(z)}{f(z)} dz + \int_{C_2} \frac{f'(z)}{f(z)} dz &= \int_{C_1} \frac{f'(z)}{f(z)} dz + \int_{-C_1} \frac{f'(-\frac{1}{z})}{f(-\frac{1}{z})} d\left(-\frac{1}{z}\right) \\
&= \int_{C_1} \frac{f'(z)}{f(z)} dz + \int_{-C_1} \frac{f'(-\frac{1}{z})}{z^2 f(-\frac{1}{z})} d(z)
\end{aligned} \tag{43}$$

Looking at the second term,

$$\begin{aligned}
\frac{d}{dz} \left(f\left(-\frac{1}{z}\right) \right) &= \frac{d}{dz} (z^k f(z)) \\
f'\left(-\frac{1}{z}\right) \frac{1}{z^2} &= kz^{k-1} f(z) + z^k f'(z) \\
f'\left(-\frac{1}{z}\right) &= kz^{k+1} f(z) + z^{k+2} f'(z) \\
\Rightarrow \frac{f'(-\frac{1}{z})}{z^2 f(-\frac{1}{z})} &= \frac{kz^{k+1} f(z) + z^{k+2} f'(z)}{z^{k+2} f(z)} \\
&= \frac{k}{z} + \frac{f'(z)}{f(z)}
\end{aligned} \tag{44}$$

$$\begin{aligned}
\int_{C_1} \frac{f'(z)}{f(z)} dz + \int_{-C_1} \frac{f'(-\frac{1}{z})}{z^2 f(-\frac{1}{z})} d(z) &= \int_{C_1} \left(\frac{f'(z)}{f(z)} - \left[\frac{k}{z} + \frac{f'(z)}{f(z)} \right] \right) dz \\
&= \int_{C_1} -\frac{k}{z} dz \xrightarrow{\varepsilon \rightarrow 0} \frac{k}{12}
\end{aligned} \tag{45}$$

Thus the arcs of unit circle give us $\frac{k}{12}$. Lets do the arcs around i and ω ,

$$\int_{C_i} \frac{f'(z)}{f(z)} dz \xrightarrow{\varepsilon \rightarrow 0} \frac{1}{2} \int_{\varepsilon \text{ counter clockwise circle around } i} \frac{f'(z)}{f(z)} dz = \frac{-1}{2} v_i(f) \tag{46}$$

$$\int_{C_\omega} \frac{f'(z)}{f(z)} dz \xrightarrow{\varepsilon \rightarrow 0} -\frac{1}{6} v_\omega(f) \tag{47}$$

$$\int_{C_{-\omega^{-1}}} \frac{f'(z)}{f(z)} dz \xrightarrow{\varepsilon \rightarrow 0} -\frac{1}{6} v_\omega(f) \tag{48}$$

Thus putting every thing together we see,

$$\lim_{\varepsilon \rightarrow 0} \int_{\Gamma_\varepsilon} \frac{f'(z)}{f(z)} dz = -\frac{1}{3} v_\omega(f) - \frac{1}{2} v_i(f) - v_\infty(f) + \frac{k}{12} \tag{49}$$

thus,

$$\frac{k}{12} = \frac{1}{3} v_\omega(f) + \frac{1}{2} v_i(f) + v_\infty(f) + \sum_{z \in \Gamma \setminus D_F} v_z(f) \tag{50}$$

□

Definition 1.4.2 $E_k : \mathbb{H} \rightarrow \mathbb{D}$ where $E_k = P_{0,k}$ is a **Eisenstein Series**

Lemma 1.4.3 For $k \geq 0$ even, $E_k \in M_k - \{0\}$ and

$$\lim_{y \rightarrow \infty} E_k(x + iy) = 1 \quad (51)$$

Proof Recall,

$$E_k(z) = \sum_{g \in \overline{T} \setminus \Gamma} j_k(g, z) = \sum_{g \in \overline{T} \setminus \Gamma} (cz + d)^{-k} \quad (52)$$

for $z \in \mathbb{H}$. Since E_k is holomorphic on $\mathbb{H}$, $\widetilde{E}_k : \mathbb{D}^\times \rightarrow \mathbb{C}$ is holomorphic. To prove the lemma, it suffices to prove that $\lim_{y \rightarrow \infty} E_k(x + iy) = 1$ by the removable singularity theorem.

Note that $\forall g \in \overline{T}, j_k(g, z) = 1. \forall g \in \Gamma - \overline{T}, \forall z = x + iy \in D_F \Rightarrow |x| \leq \frac{1}{2}$

$$\begin{aligned} |c(x + iy) + d|^2 &= |(cx + d) + icy| \\ &= (cx + d)^2 + c^2 y^2 \\ &= c^2 x^2 + 2cdx + d^2 + c^2 y^2 \\ &\geq c^2 x^2 - cd + d^2 + c^2 y^2 \\ &= c^2(x^2 + y^2) - cd + d^2 \\ &\geq c^2 - cd + d^2 \end{aligned} \quad (53)$$

$x^2 + y^2 = 1$ corresponds to the case where $z = e^{i\theta}$, but since $x = -\frac{1}{2} \Rightarrow z = e(\frac{1}{3})$.

Thus

$$|c(x + iy) + d|^2 \geq \left| ce\left(\frac{1}{3}\right) + d \right|^2 \quad (54)$$

.

And we know that, $\sum_{(c,d) \in \mathbb{Z}^2 - (0,0)} |ce(\frac{1}{3}) + d|^{-k}$ converges for $k \geq 4$, by Lemma 1.2.8. Thus the convergence of E_0 is dominated, thus by dominated convergence theorem you can interchange limits

$$\begin{aligned} \lim_{y \rightarrow \infty} \sum_{g \in \overline{T} \setminus \Gamma} j_k(g, x + iy) &= \sum_{g \in \overline{T} \setminus \Gamma} \lim_{y \rightarrow \infty} j_k(g, x + iy) \\ &= 1 \quad \text{for } g = I \end{aligned} \quad (55)$$

From this we know that $\widetilde{E}_k$ is bounded around a region of 0, Since its meromorphic, it follows that E_k is holomorphic at 0, so $E_k \in M_k$. Since $E_k(\infty) = 1 \Rightarrow E_k \notin S_k$. $\square$

Theorem 1.4.4 $M_k = \{0\}$ for $k < 0$, k odd, and $k = 2$.

Proof As previously proven, $M_k = \{0\}$ for k odd, since $\forall f$ weight k modular forms, $f(-Iz) = f(z) = (-1)^k f(z) \Rightarrow f$ is zero or k is even.

If $f \in M_{2k} - \{0\}$, by Theorem 1.4.1, $\exists a, b, c \geq 0$ such that

$$\frac{k}{6} = \frac{2k}{12} = \frac{a}{3} + \frac{b}{2} + c \quad (56)$$

1. $k < 0$: There are no positive integers that sum to negative numbers. Thus by contradiction $M_{2k} - \{0\} = \emptyset$.
2. $k = 1$: $\frac{a}{3} + \frac{b}{2} + c > \frac{1}{3}$ for all $a, b, c \geq 0$. Thus by contradiction $M_{2k} = \{0\}$

□

Proposition 1.4.5 $\exists \Delta \in S_{12} - \{0\}$ such that $v_z(\Delta) = 0$ for $z \in \mathbb{H}$.

Proof Let $\Delta = E_4^3 - E_6^2$. $\forall g \in \text{SL}_2(\mathbb{Z})$, $(\Delta|_{12}g) = (E_4^3)|_{12} - (E_6^2)|_{12} = E_3^4 - E_6^2 = \Delta$. Thus $\Delta \in M_{12}$. By Lemma 1.4.3,

$$\lim_{y \rightarrow \infty} \Delta(x + iy) = \lim_{y \rightarrow \infty} (E_4^3 - E_6^2)(x + iy) = 1 - 1 = 0 \quad (57)$$

Thus $\Delta \in S_{12}$. We need to show tho that $\Delta \neq 0$. For this look at Theorem 1.4.1, for E_4 and E_6

$$\frac{1}{3} = \frac{v_i(E_4)}{2} + \frac{v_\omega(E_4)}{3} + \sum_{z \in \Gamma \backslash \mathbb{H}} v_z(E_4) + v_\infty(E_4) \quad (58)$$

which implies

$$v_z(E_4) = \begin{cases} 1 & \text{if } z = \omega \\ 0 & \text{otherwise} \end{cases} \quad (59)$$

For E_6 we see,

$$\frac{1}{2} = \frac{v_i(E_6)}{2} + \frac{v_\omega(E_6)}{3} + \sum_{z \in \Gamma \backslash \mathbb{H}} v_z(E_6) + v_\infty(E_6) \quad (60)$$

thus,

$$v_z(E_6) = \begin{cases} 1 & \text{if } z = i \\ 0 & \text{otherwise} \end{cases} \quad (61)$$

Thus $\Delta(i) \neq 0 \Rightarrow \Delta \neq 0$. □

Theorem 1.4.6 $\dim S_k = \dim M_k$ or $\dim M_k - 1$

Proof Let $A : M_k \rightarrow \mathbb{C}$, where $f \mapsto \tilde{f}(0)$. $S_k = \ker A$. A is $\mathbb{C}$ -linear. Thus by rank nullity, $\dim S_k = M_k - \dim A(M_k)$, where $\dim A(M_k) \leq 1$. Thus $\dim S_k = M_k$ or $M_k - 1$. □

Theorem 1.4.7 For all $k \in \mathbb{N}$, let $u : M_{2k} \rightarrow S_{2k+12}$ where $f \mapsto \Delta f$. u is an isomorphism of vector spaces.

Proof

1. **Linearity**: $\forall (f, g, a, b) \in M_{2k}^2 \times \mathbb{C}^2$, $u(af + bg) = \Delta \times (af + bg) = a\Delta f + b\Delta g$.
2. **Injectivity**: $\forall f, g \in M_{2k} \ni \Delta f = \Delta g$
3. **im**(u) $\subset S_{2k+12}$: $(f\Delta)(\infty) = 0$

Let $f \in S_{2k+12}$, $g := \frac{f}{\Delta}$. Thus g is modular of weight $2k$. g is holomorphic everywhere, because $v_\infty(g) = v_\infty(f) - 1 \geq 0$ and $\forall z \in \mathbb{H}$, $v_z(g) = v_z(f) \geq 0$ (since Δ has no zeros in $\mathbb{H}$). Thus $g \in M_{2k}$. Thus u is a surjective map and thus a linear isomorphism from $M_{2k} \rightarrow S_{2k+12}$.

By Theorem 1.4.4, $M_{2k} = \{0\}$ for $k < 0$ and $k = 1$. Thus $S_{2k+12} = 0$. Thus $S_0, S_4, S_6, S_8, S_{10} = \{0\}$. Since $1, E_4, E_6, E_8, E_{10} \in M_k - \{0\}$ (Lemma 1.4.3), and $\dim M_k = \dim S_k$ or $\dim S_k + 1 \Rightarrow 1, E_4, E_6, E_8, E_{10}$ generate $M_0, M_4, M_6, M_8, M_{10}$. Thus since $S_{12} \cong M_0 \Rightarrow S_{12}$ **generated by Δ** . $\square$

Theorem 1.4.8 (*Dimension of Modular Forms*) For $k \geq 1$

$$\dim M_{2k} = \begin{cases} \left\lfloor \frac{k}{6} \right\rfloor & \text{if } k \equiv 1 \pmod{6} \\ \left\lfloor \frac{k}{6} \right\rfloor + 1 & \text{otherwise} \end{cases} \quad (62)$$

Proof Observe that for $2k < 12$, that the theorem holds.

1. For $2k = 12$, note that $E_{12} \notin S_{12} \Rightarrow M_{12}$ generated by E_{12} and Δ . Thus $\dim M_{12} = 2$.
2. For $2k = 14$, note that $E_{14} \in M_{14} - S_{14}$ (Lemma 1.4.3) and $S_{14} = 0$. By Theorem 1.4.4, $\dim M_{14} = 1$.

Suppose the formula holds for k . Then $\dim S_{2(k+6)} = \dim M_{2k} \Rightarrow \dim M_{2(k+6)} = \dim S_{2(k+6)} + 1 = \dim M_{2k} + 1$.

$$\begin{aligned} \dim M_{2(k+6)} &= \begin{cases} \left\lfloor \frac{k}{6} \right\rfloor + 1 & \text{if } k \equiv 1 \pmod{6} \\ \left\lfloor \frac{k}{6} \right\rfloor + 2 & \text{otherwise} \end{cases} \\ &= \begin{cases} \left\lfloor \frac{k+6}{6} \right\rfloor & \text{if } k \equiv 1 \pmod{6} \\ \left\lfloor \frac{k+6}{6} \right\rfloor + 1 & \text{otherwise} \end{cases} \\ &= \begin{cases} \left\lfloor \frac{k+6}{6} \right\rfloor & \text{if } (k+6) \equiv 1 \pmod{6} \\ \left\lfloor \frac{k+6}{6} \right\rfloor + 1 & \text{otherwise} \end{cases} \end{aligned} \quad (63)$$

Thus by induction the formula holds for all k . $\square$

Corollary 1.4.9 The $\mathbb{C}$ -algebra

$$\bigoplus_{k \geq 0} M_k \quad (64)$$

with product of modular forms as multiplication is $\cong$ to $\mathbb{C}[x, y]$ by $u : \mathbb{C}[x, y] \rightarrow \bigoplus_{k \geq 0} M_k$ where

$$\begin{aligned} x &\rightarrow E_4 \\ y &\rightarrow E_6 \end{aligned} \quad (65)$$

Proof $\forall f \in \ker(u)$, $f = \sum_{i,j} a_{i,j} x^i y^j$. Thus $u(f) = \sum_{i,j} a_{i,j} E_4^i E_6^j = 0$ is a relation. Then for all k , each peice of weight k vanishes. Thus $\sum_{4i+6j=k} a_{i,j} E_4^i E_6^j = 0$.

Pick $(a, b) \in \mathbb{N}$ with $4a + 6b = k$ and a minimal, we observe that for all i, j with $4i + 6j = k$ we have $i \geq a \Rightarrow j \leq b$.

$$\frac{E_4^a E_6^b}{E_4^i E_6^j} = \frac{E_6^{b-j}}{E_4^{i-a}} \quad (66)$$

Since $4(i-a) = 6(b-j) \Rightarrow b-j = \frac{2}{3}(i-a)$,

$$\begin{aligned} \frac{E_4^a E_6^b}{E_4^i E_6^j} &= \frac{E_6^{b-j}}{E_4^{i-a}} \\ &= \frac{E_6^{\frac{2}{3}(i-a)}}{E_4^{i-a}} \\ &= \left(\frac{E_6^2}{E_4^3} \right)^{\frac{i-a}{3}} \end{aligned} \quad (67)$$

Thus

$$E_4^a E_6^b = E_4^i E_6^j \left(\frac{E_6^2}{E_4^3} \right)^{\frac{i-a}{3}} \quad (68)$$

Thus,

$$\begin{aligned} 0 &= \sum_{4i+6j} a_{i,j} E_4^i E_6^j = \sum_{4i+6j} a_{i,j} E_4^a E_6^b \left(\frac{E_4^3}{E_6^2} \right)^{\frac{i-a}{3}} \\ &= E_4^a E_6^b \sum_{4i+6j} a_{i,j} \left(\frac{E_4^3}{E_6^2} \right)^{\frac{i-a}{3}} \\ &\Rightarrow \sum_{4i+6j} a_{i,j} \left(\frac{E_4^3}{E_6^2} \right)^{\frac{i-a}{3}} = 0 \end{aligned} \quad (69)$$

This is only possible in two cases:

1. $a_{i,j} = 0$: Thus $f = 0$.
2. $\left(\frac{E_4^3}{E_6^2} \right)^{\frac{i-a}{3}} = c$: Not true since E_4 and E_6 don't agree on i and $e(\frac{1}{3})$.

Thus $f = 0 \Rightarrow u$ is injective.

u is surjective: Induction on k that $M_k \subset \text{im}(u)$. This holds for $k \leq 6$ since $M_0 = \mathbb{C}$, $M_2 = 0$, $M_4 = \mathbb{C} \cdot E_4$, and $M_6 = \mathbb{C} \cdot E_6$.

Suppose it hold for all $n < k$ that $M_n \subset \text{im}(u)$. Then pick $a, b \ni 4a + 6b = k$. $\forall f \in M_6$, there $\exists \lambda \ni f - \lambda E_4^a E_6^b \in S_k$, because E_4, E_6 are not cusp forms $E_4^a E_6^b$ is not one either so its value at infinity is nonzero. Thus $f = E_4^a E_6^b + \Delta g$ for $g \in M_{k-12}$. Since $\Delta \in \text{im}(u)$, by inductive hypothesis, $f \in \text{im}(u)$. $\square$

1.5. The J function

Corollary 1.5.1 (*The J function*) There exists a meromorphic modular form $J : \mathbb{H} \rightarrow \mathbb{C}$ of weight 0 such that J defines a bijection between $\mathrm{SL}_2(\mathbb{Z})/\mathbb{H}$ and $\mathbb{C}$ and J has a pole at ∞ and residue 1.

Proof Since Δ has a simple zero at ∞ and no other zero, we can find a $\lambda \in \mathbb{C}^\times$ such that $J = \lambda \frac{E_4^3}{\Delta}$ has a simple pole with residue 1 at ∞ . It is weight 0, and holomorphic on $\mathbb{H}$. J defines a map $\mathrm{SL}_2(\mathbb{Z})/\mathbb{H} \rightarrow \mathbb{C}$. To prove it is a bijection, pick $z \in \mathbb{C}$. The function $f_z : \lambda E_4^3 - z\Delta$ is in M_{12} and is nonzero (E_4 is not a cusp form). The formula of zeros from Theorem 1.4.1 gives,

$$v_\infty(f_z) + \frac{1}{2}v_i(f_z) + \frac{1}{3}v_{e(\frac{1}{3})}(f_z) + \sum_{w \in \mathrm{SL}_2(\mathbb{Z})/\mathbb{H} - \{i, e(\frac{1}{3})\}} v_w(f_z) = 1 \quad (70)$$

Note $v_\infty(f_z) = 0$. Thus there are 3 cases:

1. $v_i(f_z) = 2$ and all others are 0
2. $v_{e(\frac{1}{3})}(f_z) = 3$ and all others are 0
3. $v_w(f_z) = 1$ for $w \neq i, e(\frac{1}{3})$ and all others are 0.

In any of the 3 cases, f_z has unique zero $w \in \mathrm{SL}_2(\mathbb{Z})/\mathbb{H}$. Since $(\lambda E_4^3 - z\Delta)(w) = f_z(w) = 0 \Rightarrow j(w) = z$ where w is the unique solution. Thus J is injective and surjective. Thus J is a meromorphic bijection. $\square$

Corollary 1.5.2 Any modular function f which is meromorphic everywhere, including at ∞ , is of the form

$$f = Q(j) \quad (71)$$

for $Q \in \mathbb{C}(x)$, (field of rational functions over $\mathbb{Q}$).

Proof

1. WLOG we can assume that f is holomorphic on $\mathbb{H}$: Examine f on $\mathrm{SL}_2(\mathbb{Z})/\mathbb{H}$.
 1. $\tilde{f}$ has a pole at 0: Since poles are isolated then $\exists 0 < r_0 < 1$ such that $\tilde{f}$ only has 1 pole on $\{w \in \mathbb{D} \mid |w| < r\}$ which is 0. Thus $\exists y_0 > 0$ such that $\forall z \in \mathrm{SL}_2(\mathbb{Z})/\mathbb{H}, \Im(z) > y_0, f$ has no pole. Let $K = \mathrm{SL}_2(\mathbb{Z})/\mathbb{H} \cap \{z \in \mathbb{H} \mid \Im(z) \leq y_0\}$ which is a compact subspace. Let S be the set of poles of f , which is a discrete subspace of $\mathrm{SL}_2(\mathbb{Z})/\mathbb{H}$. $S \cap K$ must be finite set (since $S \cap K$ would have a accumulation point in K contradicting the discreteness of S), thus f has a finite number of poles in $\mathrm{SL}_2(\mathbb{Z})/\mathbb{H}$.
 2. f doesn't have a pole at 0: Basically same argument, but now you have a r_0 such that $\{w \in \mathbb{D} \mid |w| < r_0\}$ has no poles.

Let $\alpha_1, \dots, \alpha_m$ be the poles of f . Then

$$g = f \prod_{i=1}^m (j - j(\alpha_i))^{-v_{\alpha_i}(f)} \quad (72)$$

is a modular form which is holomorphic on $\mathbb{H}$.

2. Since Δ has a simple zero at ∞ , $\exists n \geq 0 \ni g = \Delta^n f$ is holomorphic modular form of weight 12, that is holomorphic at ∞ . Thus $g \in M_{12}$. Thus by Corollary 1.4.9,

$$g = \sum_{4i+6j=12n} a_{i,j} E_4^i E_6^j \quad (73)$$

For all $i, j \ni 4i + 6j = 12n \Rightarrow 4i = 3(4n - 2j)$ since $3 \nmid 4 \Rightarrow 3 \mid i$. $6j = 4(3n - i) \Rightarrow 3j = 2(3n - i) \Rightarrow 2 \mid j$. Then

$$\begin{aligned} f &= \frac{g}{\Delta^n} = \sum_{i,j} a_{i,j} \frac{(E_4^3)^{\frac{i}{3}} (E_6^2)^{\frac{j}{2}}}{\Delta^{\frac{4}{12}i + \frac{6}{12}j}} \\ &= \sum_{i,j} a_{i,j} \frac{(E_4^3)^{\frac{i}{3}} (E_6^2)^{\frac{j}{2}}}{\Delta^{\frac{i}{3} + \frac{j}{2}}} = \sum_{i,j} a_{i,j} \frac{(E_4^3)^{\frac{i}{3}}}{\Delta^{\frac{i}{3}}} \frac{(E_6^2)^{\frac{j}{2}}}{\Delta^{\frac{j}{2}}} \end{aligned} \quad (74)$$

Thus it suffices to check that $\frac{E_4^3}{\Delta}$ and $\frac{E_6^2}{\Delta}$ are rational functions of j .

$$j := \lambda \frac{E_4^3}{\Delta} \Rightarrow \frac{E_4^3}{\Delta} = \frac{j}{\lambda} \quad (75)$$

and

$$\frac{E_6^2}{\Delta} = \frac{E_4^3 - \Delta}{\Delta} = \frac{j}{\lambda} - 1 \quad (76)$$

Thus f is a rational function of j □

Remark 1.5.3 This property that all modular forms which are meromorphic everywhere is analogous to the fact that all meromorphic $f : \mathbb{C} \rightarrow \mathbb{C}$ are rational functions.

1.6. Fourier Expansions of Eisenstein and Poincaré Series

Definition 1.6.1 For $\Re(s) > 0$,

$$\Gamma(s) = \int_0^\infty t^s e^{-t} \frac{dt}{t} \quad (77)$$

Theorem 1.6.2 $\forall k \geq 4, m \geq 1$ integers. The fourier expansion of $P_{m,k}$ is given by:

$$P_{m,k}(z) = \sum_{n \geq 1} p(m, n) e(nz) \quad (78)$$

where

$$p(m, n) := \delta(m, n) + \left(\frac{m}{n}\right)^{\frac{k-1}{2}} \frac{2\pi}{i^k} \sum_{c \geq 1} \frac{1}{c} S(m, n; c) J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c}\right) \quad (79)$$

where

$$S(m, n; c) = \sum_{x \in \mathbb{Z}_c^\times} e\left(\frac{mx + n\bar{x}}{c}\right) \quad (80)$$

is the Kloosterman Sum, and

$$J_\nu(\xi) = \sum_{n \geq 0} \frac{(-1)^n}{n! \Gamma(n+1+\nu)} \left(\frac{\xi}{2}\right)^{\nu+2n} \quad (81)$$

is the Bessel Function.

Proof Recall

$$P_{m,k}(z) = \sum_{g \in \overline{T}/\mathrm{SL}_2(\mathbb{Z})} j_k(g, z) e(mgz) = \sum_{g \in \overline{T}/\mathrm{SL}_2(\mathbb{Z})} (e_m |_k g)(z) \quad (82)$$

Thus, by Corollary 5.2.2,

$$\begin{aligned} P_{m,k}(z) &= e_m(z) + \sum_{c \geq 1} \sum_{d \in \mathbb{Z}_c^\times} \sum_{n \in \mathbb{Z}} \frac{e_m\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}(z+n)\right)}{(c(z+n) + d)^k} \\ &= e(mz) + \sum_{c \geq 1} \sum_{d \in \mathbb{Z}_c^\times} \sum_{n \in \mathbb{Z}} \frac{e\left(m\begin{pmatrix} a & b \\ c & d \end{pmatrix}(z+n)\right)}{(c(z+n) + d)^k} \end{aligned} \quad (83)$$

Note:

$$\begin{aligned}
\begin{pmatrix} a & b \\ c & d \end{pmatrix}(z+n) &= \frac{a(z+n)+b}{c(z+n)+d} \\
&= \frac{ca(z+n)+bc}{c^2(z+n)+dc} \\
&= \frac{ca(z+n)+ad-1}{c^2(z+n)+cd} \\
&= \frac{a(c(z+n)+d)}{c(c(z+n)+d)} - \frac{1}{c(cz+cn+d)} \\
&= \frac{a}{c} - \frac{1}{c(cz+cn+d)}
\end{aligned} \tag{84}$$

Thus,

$$\begin{aligned}
P_{m,k}(z) &= e(mz) + \sum_{c \geq 1} \sum_{d \in \mathbb{Z}_c^\times} \sum_{n \in \mathbb{Z}} \frac{e\left(m\left(\frac{a}{c} - \frac{1}{c(cz+cn+d)}\right)\right)}{(c(z+n)+d)^k} \\
&= e(mz) + \sum_{c \geq 1} \sum_{d \in \mathbb{Z}_c^\times} e\left(\frac{ma}{c}\right) \sum_{n \in \mathbb{Z}} \frac{e\left(-\frac{m}{c(cz+cn+d)}\right)}{(c(z+n)+d)^k}
\end{aligned} \tag{85}$$

For fixed c, d , let our test function be $\varphi(x) = \frac{1}{(c(z+x)+d)^k} e\left(-\frac{m}{c(cz+cx+d)}\right)$. Note the following:

1. $c(z+x)+d \neq 0$ for all $x \in \mathbb{R}$ because $\Im(z) > 0$. Thus $\varphi \in C^\infty(\mathbb{R})$.
2. $\varphi \in L^1(\mathbb{R})$: This is because $|e(-m(\dots))| \leq 1$ and $k \geq 4$.
3. $\sum_{n \in \mathbb{Z}} \varphi(x+n)$ is locally absolutely convergent.
4. $\varphi'(x) = -kc(c(z+x)+d)^{-k-1}e(\dots) + (c(z+x)+d)^{-k} \frac{2\pi im}{c}(cz+cx+d)^{-2}e(\dots)$.

Thus φ' is locally absolutely convergent for a similarly reason to φ .

By Theorem 5.1.1 in the Appendix,

$$\sum_{n \in \mathbb{Z}} \frac{e\left(-\frac{m}{c(cz+cn+d)}\right)}{(c(z+n)+d)^k} = \sum_{h \in \mathbb{Z}} \hat{\varphi}(h) \tag{86}$$

where,

$$\begin{aligned}
\hat{\varphi}(h) &= \int_{\mathbb{R}} \frac{e\left(-\frac{m}{c(cz+ct+d)}\right)}{(c(z+t)+d)^k} e(-ht) dt \\
&= \int_{\mathbb{R}} \frac{1}{(c(z+t)+d)^k} e\left(-\frac{m}{c(cz+ct+d)} - ht\right) dt \\
&= \int_{\mathbb{R}} \frac{1}{(c(x+iy+t)+d)^k} e\left(-\frac{m}{c(cx+icy+ct+d)} - ht\right) dt \quad (z = x + iy) \\
&= \int_{\mathbb{R}} \frac{1}{\left(c(x+iy+t) + \frac{c(d)}{c}\right)^k} e\left(-\frac{m}{c\left(cx+icy+ct + \frac{c(d)}{c}\right)} - ht\right) dt \\
&= \int_{\mathbb{R}} \frac{1}{\left(c\left(x+iy+t + \frac{d}{c}\right)\right)^k} e\left(-\frac{m}{c^2\left(x+iy+t + \frac{d}{c}\right)} - ht\right) dt \quad (87) \\
&= \int_{\mathbb{R}} \frac{1}{(c(u+iy))^k} e\left(-\frac{m}{c^2(u+iy)} - h\left(u-x-\frac{d}{c}\right)\right) du \quad \left(u = x+t+\frac{d}{c}\right) \\
&= e\left(h\left(\frac{d}{c}\right)\right) \int_{\mathbb{R}} \frac{1}{(c(u+iy))^k} e\left(-\frac{m}{c^2(u+iy)} - h(u-x)\right) du \\
&= e\left(h\left(\frac{d}{c}\right)\right) \int_{\mathbb{R}} \frac{1}{(c(u+iy))^k} e\left(-\frac{m}{c^2(u+iy)} - h(u-z+iy)\right) du \\
&= e\left(h\left(z+\frac{d}{c}\right)\right) \int_{\mathbb{R}} \frac{1}{(c(u+iy))^k} e\left(-\frac{m}{c^2(u+iy)} - h(u+iy)\right) du \\
&= e\left(h\left(z+\frac{d}{c}\right)\right) \int_{\mathfrak{I}(w)=y} \frac{1}{(cw)^k} e\left(-\frac{m}{c^2w} - hw\right) dw
\end{aligned}$$

Thus,

$$\sum_{n \in \mathbb{Z}} \frac{e\left(-\frac{m}{c(cz+cn+d)}\right)}{(c(z+n)+d)^k} = \sum_{h \in \mathbb{Z}} e\left(h\left(z+\frac{d}{c}\right)\right) \int_{\mathfrak{I}(w)=y} \frac{1}{(cw)^k} e\left(-\frac{m}{c^2w} - hw\right) dw \quad (88)$$

Plugging in Equation 83,

$$\begin{aligned}
P_{m,k}(z) &= e(mz) + \sum_{c \geq 1} \sum_{d \in \mathbb{Z}_c^\times} e\left(\frac{ma}{c}\right) \sum_{h \in \mathbb{Z}} e\left(h\left(z+\frac{d}{c}\right)\right) \int_{\mathfrak{I}(w)=y} \frac{1}{(cw)^k} e\left(-\frac{m}{c^2w} - hw\right) dw \\
&= e(mz) + \sum_{c \geq 1} \sum_{d \in \mathbb{Z}_c^\times} \sum_{h \in \mathbb{Z}} e\left(\frac{ma+hd}{c}\right) \left(\int_{\mathfrak{I}(w)=y} \frac{1}{(cw)^k} e\left(-\frac{m}{c^2w} - hw\right) dw \right) e(hz) \quad (89) \\
&= e(mz) + \sum_{h \in \mathbb{Z}} \sum_{c \geq 1} \sum_{d \in \mathbb{Z}_c^\times} e\left(\frac{ma+hd}{c}\right) \left(\int_{\mathfrak{I}(w)=y} \frac{1}{(cw)^k} e\left(-\frac{m}{c^2w} - hw\right) dw \right) e(hz) \\
&= e(mz) + \sum_{h \in \mathbb{Z}} p_k(m, h) e(hz)
\end{aligned}$$

where,

$$\begin{aligned}
p_k(m, h) &:= \sum_{c \geq 1} \sum_{d \in \mathbb{Z}_c^\times} e\left(\frac{ma + hd}{c}\right) \left(\int_{\mathfrak{I}(w)=y} \frac{1}{(cw)^k} e\left(-\frac{m}{c^2 w} - hw\right) dw \right) \\
&= \sum_{c \geq 1} \left(\int_{\mathfrak{I}(w)=y} \frac{1}{(cw)^k} e\left(-\frac{m}{c^2 w} - hw\right) dw \right) \sum_{d \in \mathbb{Z}_c^\times} e\left(\frac{ma + hd}{c}\right)
\end{aligned} \tag{90}$$

Note that since $ad - bc = 1 \Rightarrow ad = 1 + bc \Rightarrow ad \equiv 1 \pmod{c} \Rightarrow a \equiv \bar{d} \pmod{c}$. Thus,

$$\sum_{d \in \mathbb{Z}_c^\times} e\left(\frac{ma + hd}{c}\right) = \sum_{d \in \mathbb{Z}_c^\times} e\left(\frac{m\bar{d} + hd}{c}\right) = S(m, h; c) \tag{91}$$

Note $S(m, h; c) = S(h, m; c)$ substitute d with $|d)$.

Thus

$$p_k(m, h) = \sum_{c \geq 1} \left(\int_{\mathfrak{I}(w)=y} \frac{1}{(cw)^k} e\left(-\frac{m}{c^2 w} - hw\right) dw \right) S(m, h; c) \tag{92}$$

Thus all that is left is to handle the complex integral. Let

$$I_k(m, h; c) = \int_{\mathfrak{I}(w)=y} \frac{1}{(cw)^k} e\left(-\frac{m}{c^2 w} - hw\right) dw \tag{93}$$

1. $I_k(m, k; c)$ invariant under choice of y :

$$I_{k(m,k;c)} = \lim_{T \rightarrow \infty} \int_{-T}^T \frac{1}{(c(x+iy))^k} e\left(-\frac{m}{c^2(x+iy)} - h(x+iy)\right) dx \quad (94)$$

$\forall y' > 0$, let $C_{y'}$ be the box contour with $[-T+iy, T+iy]$ and $[T+iy', -T+iy']$ as top and bottom (flipped signs for orientation.) Since the integrand is holomorphic on the interior, by Cauchy's theorem:

$$\int_{C_{y'}} \frac{1}{(c(x+iy))^k} e\left(-\frac{m}{c^2(x+iy)} - h(x+iy)\right) dx = 0 \quad (95)$$

Vertical segments suffices to just see 1 side:

$$\begin{aligned} \left| e\left(-\frac{m}{c^2(T+iu)} - h(T+iu)\right) \right| &= \left| e\left(-\frac{m}{c^2(T+iu)}\right) \right| |e(-h(T+iu))| \\ &= \left| e\left(-\frac{mT - ium}{c^2|T+iu|^2}\right) \right| |e(-ihu)| \\ &= \left| e\left(\frac{ium}{c^2|T+iu|^2}\right) \right| |e^{2\pi hu}| \\ &\leq e^{2\pi hu} \end{aligned} \quad (96)$$

Thus,

$$\begin{aligned} &\left| \int_y^{y'} \frac{1}{(c(T+iu))^k} e\left(-\frac{m}{c^2(T+iu)} - h(T+iu)\right) du \right| \\ &\leq \int_y^{y'} \left| \frac{1}{(c(T+iu))^k} e\left(-\frac{m}{c^2(T+iu)} - h(T+iu)\right) \right| du \leq \int_y^{y'} \frac{1}{c^k|T+iu|^k} e^{2\pi hu} du \xrightarrow{T \rightarrow \infty} 0 \end{aligned} \quad (97)$$

The same with the opposite vertical. Thus

$$\int_{\mathfrak{I}(w)=y} (cw)^{-k} e\left(-\frac{m}{c^2w} - hw\right) dx = \int_{\mathfrak{I}(w)=y'} (cw)^{-k} e\left(-\frac{m}{c^2w} - hw\right) dx \quad (98)$$

2. If $h \leq 0$, then $I_k(m, h; c) = 0$: Let $h \geq 0$, then

$$\begin{aligned} |I_k(m, -h; c)| &\leq \int_{\mathbb{R}} \left| \frac{1}{(c(t+iy))^k} e\left(-\frac{m}{c^2(t+iy)} + h(t+iy)\right) \right| dw \\ &= \int_{\mathbb{R}} (c|t+iy|)^{-k} \left| e\left(-\frac{m}{c^2(t+iy)} + h(t+iy)\right) \right| dw \\ &= \int_{\mathbb{R}} (c|t+iy|)^{-k} \left| e\left(\frac{imy - mx}{c^2|t+iy|^2} + h(t+iy)\right) \right| dw \\ &\leq \int_{\mathbb{R}} (c|t+iy|)^{-k} \left| e\left(\frac{imy}{c^2|t+iy|^2}\right) \right| |e(h(t+iy))| dw \\ &\leq \int_{\mathbb{R}} (c|t+iy|)^{-k} |e(hiy)| dw \\ &= \int_{\mathbb{R}} (c|t+iy|)^{-k} e^{-hy} dw \end{aligned} \quad (99)$$

Thus,

$$P_{m,k}(z) = e(mz) + \sum_{h \geq 1} p_k(m, h) e(hz) \quad (100)$$

Consequences: $P_{m,k}$ is holomorphic at ∞ for $m \geq 0$. Furthermore, since

$$|P_{m,k}(x + iy)| \leq e^{-my} + \sum_{h \geq 0} |p_k(m, h)| |e(-hy)| \quad (101)$$

as $y \rightarrow \infty$ for $m \geq 1$ $P_{m,k}(x + iy) \rightarrow 0$. Thus $P_{m,k} \in S_k$.

Continuing on,

3. Computing $I_k(m, h; c)$:

1. $m = 0$ and $h \geq 1$: Then

$$\begin{aligned}
 I(0, h; c) &= \int_{\Im(w)=y} (cw)^{-k} e(-hw) dw \\
 &= \frac{1}{h} \int_{\Im(w)=y} \left(\frac{cv}{h}\right)^{-k} e(-v) dv \\
 &= c^{-k} h^{k-1} \int_{\Im(w)=y} v^{-k} e(-v) dv \\
 &= \frac{h^{k-1} (2\pi)^k}{c^k i^k \Gamma(k)}
 \end{aligned} \tag{102}$$

The last equality comes from Proposition 5.3.1 in the Appendix.

2. $m \geq 0$ and $h \geq 1$:

Expand $e\left(-\frac{m}{c^2 w}\right)$ in a power series:

$$e\left(-\frac{m}{c^2 w}\right) = \sum_{l \geq 0} \frac{1}{l!} \left(\frac{-m}{c^2 w}\right)^l (2i\pi)^l \tag{103}$$

Thus,

$$\begin{aligned}
 I(m, h; c) &= \int_{\Im(w)=y} \frac{1}{(cw)^k} e\left(-\frac{m}{c^2 w} - hw\right) dw \\
 &= \int_{\Im(w)=y} \sum_{l \geq 0} \frac{1}{l!} \frac{1}{(cw)^k} \left(\frac{-m}{c^2 w}\right)^l (2i\pi)^l e(-hw) dw \\
 &= \sum_{l \geq 0} \frac{(-2i\pi m)^l}{l!} \int_{\Im(w)=y} \frac{1}{c^{k+2l} w^{k+l}} e(-hw) dw \\
 &= \sum_{l \geq 0} \frac{(-2i\pi m)^l h^{k+l-1} (2\pi)^{k+l}}{c^{k+2l} l!} \frac{1}{i^{k+l} \Gamma(k+l)} \\
 &= \sum_{l \geq 0} \frac{(-1)^l}{l!} \left(\frac{2\pi}{c}\right)^{k+2l} \frac{h^{k+l-1} m^l}{i^k} \frac{1}{\Gamma(k-1+l+1)} \\
 &= \sum_{l \geq 0} \frac{(-1)^l}{l!} \left(\frac{4\pi}{2c}\right)^{k+2l} \frac{h^{k+l-1} m^l}{i^k} \frac{1}{\Gamma(k-1+l+1)} \\
 &= \frac{2\pi}{c i^k} \sum_{l \geq 0} \frac{(-1)^l}{l!} \left(\frac{4\pi}{2c}\right)^{k+2l-1} h^{\frac{k+2l-1}{2} + \frac{k-1}{2}} m^{\frac{k+2l-1}{2} - \frac{k-1}{2}} \frac{1}{\Gamma(k-1+l+1)} \\
 &= \frac{2\pi}{c i^k} \sum_{l \geq 0} \frac{(-1)^l}{l!} \left(\frac{4\pi\sqrt{hm}}{2c}\right)^{k+2l-1} h^{\frac{k-1}{2}} m^{-\frac{k-1}{2}} \frac{1}{\Gamma(k-1+l+1)} \\
 &= \frac{2\pi}{c i^k} \sum_{l \geq 0} \frac{(-1)^l}{l!} \left(\frac{4\pi\sqrt{hm}}{2c}\right)^{k+2l-1} \left(\frac{h}{m}\right)^{\frac{k-1}{2}} \frac{1}{\Gamma(k-1+l+1)}
 \end{aligned} \tag{104}$$

Restructure the sum a bit.

$$\begin{aligned}
I(m, h; c) &= \frac{2\pi}{ci^k} \left(\frac{h}{m}\right)^{\frac{k-1}{2}} \sum_{l \geq 0} \frac{(-1)^l}{l!} \left(\frac{4\pi\sqrt{hm}}{2c}\right)^{(k-1)+2l} \frac{1}{\Gamma((k-1)+l+1)} \\
&= \frac{2\pi}{ci^k} \left(\frac{h}{m}\right)^{\frac{k-1}{2}} J_{k-1} \left(\frac{4\pi\sqrt{hm}}{c}\right)
\end{aligned} \tag{105}$$

By the definition

$$J_\nu(\xi) = \sum_{l \geq 0} \frac{(-1)^l}{l! \Gamma(\nu + 1 - l)} \left(\frac{\xi}{2}\right)^{\nu+2l} \tag{106}$$

of the Bessel Function.

Conclusion:

$$p_k(m, h) := \begin{cases} \frac{h^{k-1}(2\pi)^k}{i^k \Gamma(k)} \sum_{c \geq 1} \frac{1}{c^k} S(m, h; c) & \text{if } m = 0 \\ \left(\frac{h}{m}\right)^{\frac{k-1}{2}} \frac{2\pi}{i^k} \sum_{c \geq 1} \frac{1}{c} S(m, h; c) J_{k-1} \left(\frac{4\pi\sqrt{hm}}{c}\right) & \end{cases} \tag{107}$$

1. Eisenstein Series (ie $m = 0$):

$$\begin{aligned}
E_k(z) &= 1 + \sum_{h \geq 1} p_k(0, h) e(hz) \\
&= \frac{(2\pi)^k}{i^k \Gamma(k)} \sum_{h \geq 0} \left(\sum_{c \geq 1} \frac{h^{k-1}}{c^k} S(0, h; c) \right) e(hz) \\
&= \frac{(2\pi)^k}{i^k (k-1)!} \sum_{h \geq 0} \left(h^{k-1} \sum_{c \geq 1} \frac{1}{c^k} S(0, h; c) \right) e(hz)
\end{aligned} \tag{108}$$

2. $m \geq 1$:

$$\begin{aligned}
P_{m,k}(z) &= e(mz) + \sum_{h \geq 1} p_k(m, h) e(hz) \\
&= \sum_{h \geq 1} (p_k(m, h) + \delta(m, h)) e(hz)
\end{aligned} \tag{109}$$

□

Theorem 1.6.3 Let $k \geq 4$ be an integer. The fourier expansion of E_k , the weight k Eisenstein Series, is

$$E_k(z) = 1 + \frac{(2i\pi)^k}{\zeta(k)(k-1)!} \sum_{n \geq 1} \sigma_{k-1}(n) e(nz) \tag{110}$$

where $\sigma_{k-1}(n) = \sum_{d|n} d^{k-1}$

Proof Carrying on from Theorem 1.6.2, we need to understand $S(0, h; c)$.

By Mobius Inversion,

$$\begin{aligned}
\sum_{d \in \mathbb{Z}_c} e\left(\frac{nd}{c}\right) &= \sum_{\delta | c} \sum_{\substack{d \in \mathbb{Z}_c \\ (d, c) = \delta}} e\left(\frac{nd}{c}\right) \\
&= \sum_{\delta | c} \sum_{\substack{\delta d' \in \mathbb{Z}_c \\ (\delta d', c) = \delta}} e\left(\frac{n\delta d'}{c}\right) \\
&= \sum_{\delta | c} \sum_{\substack{d' \in \mathbb{Z}_{c/\delta} \\ (d', c/\delta) = 1}} e\left(\frac{nd'}{\frac{c}{\delta}}\right) \\
&= \sum_{\delta | c} S\left(0, n; \frac{c}{\delta}\right)
\end{aligned} \tag{111}$$

but also

$$\sum_{d \in \mathbb{Z}_c} e\left(\frac{nd}{c}\right) = \begin{cases} c & \text{if } c \mid n \\ 0 & \text{otherwise} \end{cases} \tag{112}$$

thus

$$\sum_{\delta | c} S\left(0, n; \frac{c}{\delta}\right) = c \mathbb{1}_{c|n} \tag{113}$$

By Mobius Inversion,

$$\begin{aligned}
S(0, n; c) &= \sum_{d|c} \left(\frac{c}{d}\right) \mu(d) \mathbb{1}_{\frac{c}{d}|n} \\
&= \sum_{\delta|c} \delta \mu\left(\frac{c}{\delta}\right) \mathbb{1}_{\delta|n} \\
&= \sum_{\delta|(c, n)} \delta \mu\left(\frac{c}{\delta}\right)
\end{aligned} \tag{114}$$

Thus, for $h \geq 0$

$$\begin{aligned}
h^{k-1} \sum_{c \geq 1} \frac{1}{c^k} S(0, h; c) &= h^{k-1} \sum_{c \geq 1} \frac{1}{c^k} \sum_{d|(c, h)} d \mu\left(\frac{c}{d}\right) \\
&= h^{k-1} \sum_{d|h} d \sum_{\substack{c \geq 1 \\ d|c}} \mu\left(\frac{c}{d}\right) c^{-k} \\
&= h^{k-1} \sum_{d|h} d \sum_{\alpha \geq 1} \mu(\alpha) (\alpha d)^{-k} \\
&= h^{k-1} \sum_{d|h} d^{-(k-1)} \sum_{\alpha \geq 1} \mu(\alpha) \alpha^{-k} \\
&= \frac{1}{\zeta(k)} \sum_{d|h} \left(\frac{h}{d}\right)^{k-1} \\
&= \frac{1}{\zeta(k)} \sum_{\delta|h} \delta^{k-1} \\
&= \frac{\sigma_{k-1}(h)}{\zeta(k)}
\end{aligned} \tag{115}$$

Thus plugging this into Equation 108, we get

$$\begin{aligned}
E_k(z) &= 1 + \frac{(2\pi)^k}{i^k (k-1)!} \sum_{h \geq 1} \frac{\sigma_{k-1}(h)}{\zeta(k)} e(hz) \\
&= 1 + \frac{(2\pi)^k}{i^k (k-1)! \zeta(k)} \sum_{n \geq 1} \sigma_{k-1}(n) e(nz)
\end{aligned} \tag{116}$$

□

1.7. Petersson Formula

Lemma 1.7.1 For $f \in S_k$, the integral

$$\left| \int_{D_F} |f(z)|^2 y^k \frac{dx dy}{y^2} \right| < \infty \quad (117)$$

where D_F , is the Fundamental Domain (Equation 12).

Proof Since $f \in S_k$, there is a fourier expansion

$$f(z) = \sum_{n \geq 1} a_n e(nz) \quad (118)$$

. We can get a bound on $|f(z)|$ as $z \rightarrow \infty$:

$$\begin{aligned} |f(z)| &\leq \sum_{n \geq 1} |a_n| |e(nz)| \\ &= \sum_{n \geq 1} |a_n| e^{-2\pi n y} \\ &= e^{-2\pi y} \left(|a_1| + \sum_{n \geq 2} |a_n| e^{-2\pi(n-1)y} \right) \end{aligned} \quad (119)$$

As $y \rightarrow \infty$, each term of the series gets smaller exponentially. In fact $y \rightarrow \infty$, $\sum_{n \geq 2} |a_n| e^{-2\pi(n-1)y} \rightarrow 0$ (you can get this because $f(z) \rightarrow 0$). Let $y_0 > 0$ such that $\forall y > y_0$, $\sum_{n \geq 2} |a_n| e^{-2\pi(n-1)y} < |a_1|$. Thus $\forall y > y_0$,

$$|f(z)| \leq e^{-2\pi y} (|a_1| + |a_1|) = 2|a_1| e^{-2\pi y} \quad (120)$$

Thus the function $(x, y) \rightarrow |f(x + iy)|^2 y^{k-2}$ is Lebesgue integrable. $\square$

Definition 1.7.2 The Petersson Inner Product on S_k is defined by

$$\langle f, g \rangle = \int_{D_F} f(z) \overline{g(z)} y^k \frac{dx dy}{y^2} \quad (121)$$

This makes S_k a (finite dimensional) **Hilbert Space**, since $\langle f, f \rangle = 0 \Rightarrow f = 0$ on D_F , thus $f = 0$.

Theorem 1.7.3 For $k \geq 4$, $m \geq 1$ and $\forall f \in S_k$ we have

$$\langle f, P_{m,k} \rangle = \frac{\Gamma(k-1)}{(4\pi m)^{k-1}} a_m(f) \quad (122)$$

where

$$f(z) = \sum_{n \geq 1} a_n(f) e(nz) \quad (123)$$

is the Fourier expansion of f .

Proof

$$\begin{aligned}
\langle f, P_{m,k} \rangle &= \int_{D_F} f(z) \overline{P_{m,k}} y^k \frac{dx dy}{y^2} \\
&= \int_{D_F} f(z) \left(\sum_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} \overline{j_k(g, z) e(mgz)} \right) y^k \frac{dx dy}{y^2} \\
&= \int_{D_F} \left(\sum_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} j_k(g, z) f(gz) \overline{j_k(g, z) e(mgz)} \right) y^k \frac{dx dy}{y^2} & (f(gz) = j_{-k}(g, z) f(z)) \\
&= \int_{D_F} \left(\sum_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} |j_k(g, z)|^2 f(gz) \overline{e(mgz)} \right) y^k \frac{dx dy}{y^2} \\
&= \int_{D_F} \left(\sum_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} f(gz) \overline{e(mgz)} \right) \mathfrak{I}(gz)^k \frac{dx dy}{y^2} & (\mathfrak{I}(gz) = y |j_1(g, z)|^2) \\
&= \sum_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} \int_{D_F} f(gz) \overline{e(mgz)} \mathfrak{I}(gz)^k \frac{dx dy}{y^2} \\
&= \sum_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} \int_{gD_F} f(\omega) \overline{e(m\omega)} \mathfrak{I}(\omega)^k \frac{du dv}{v^2} & (\text{Invariance of hyperbolic measure}) \\
&= \int_{\bigcup_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} gD_F} f(\omega) \overline{e(m\omega)} v^k \frac{du dv}{v^2}
\end{aligned}$$

$\bigcup_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} gD_F$ is the fundamental domain of the action of $\overline{N}$ on $\mathbb{H}$. Since $\overline{T}$ is translations,

$$\bigcup_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} gD_F = \left\{ z \in \mathbb{H} \left| |\Re(z)| \leq \frac{1}{2} \right. \right\} \quad (125)$$

Thus,

$$\begin{aligned}
\langle f, P_{m,k} \rangle &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \int_0^\infty f(u+iv) \overline{e(m(u+iv))} v^k \frac{du dv}{v^2} \\
&= \int_0^\infty v^{k-2} \int_{-\frac{1}{2}}^{\frac{1}{2}} f(u+iv) \overline{e(m(u+iv))} du dv \\
&= \int_0^\infty v^{k-2} e^{-2\pi m v} \int_{-\frac{1}{2}}^{\frac{1}{2}} f(u+iv) e(-mu) du dv \\
&= \int_0^\infty v^{k-2} e^{-2\pi m v} \int_{-\frac{1}{2}}^{\frac{1}{2}} \sum_{n \geq 1} a_n e(n(u+iv)) e(-mu) du dv \\
&= \int_0^\infty \sum_{n \geq 1} a_n v^{k-2} e^{-2\pi(m+n)v} \int_{-\frac{1}{2}}^{\frac{1}{2}} e((n-m)u) du dv \\
&= \int_0^\infty \sum_{n \geq 1} a_n v^{k-2} e^{-2\pi(m+n)v} \delta(n, m) dv \\
&= a_m \int_0^\infty v^{k-2} e^{-4\pi m v} dv \\
&= a_m \int_0^\infty \left(\frac{u}{4\pi m} \right)^{k-2} e^{-u} \frac{1}{4\pi m} du \\
&= \frac{a_m}{(4\pi m)^{k-1}} \int_0^\infty u^{k-1} e^{-u} \frac{du}{u} \\
&= \frac{a_m \Gamma(k-1)}{(4\pi m)^{k-1}}
\end{aligned} \tag{126}$$

□

Corollary 1.7.4 The Poincare Series $P_{m,k}$ for $m \geq 1$ generate S_k as a $\mathbb{C}$ – vector space.

Proof Suppose $f \in S_k$ is orthogonal to all $P_{m,k}$ for $m \geq 1$. Then by Theorem 1.7.3, we get $a_m(f) = 0$ for all $m \geq 1$. So the space generated by $\{P_{m,k} \mid m \geq 1\}$ must be S_k . □

Corollary 1.7.5 (Petersson Formula) Let $k \geq 4$. Let $\mathcal{F}$ be an orthonormal basis of S_k . Then for $m \geq 1$ and $n \geq 1$:

$$\frac{\Gamma(k-1)}{(4\pi\sqrt{mn})^{k-1}} \sum_{f \in \mathcal{F}} a_f(m) \overline{a_f(n)} = \delta(m, n) + \frac{2\pi}{i^k} \sum_{c \geq 1} \frac{1}{c} S(m, n; c) J_{k-1} \left(\frac{4\pi\sqrt{mn}}{c} \right) \tag{127}$$

Proof Expand $P_{m,k}$ in terms of the basis $\mathcal{F}$. Thus by orthonormality of $\mathcal{F}$:

$$\begin{aligned}
P_{m,k} &= \sum_{f \in \mathcal{F}} \langle P_{m,k}, f \rangle f \\
&= \sum_{f \in \mathcal{F}} \overline{\langle f, P_{m,k} \rangle} f \\
&= \frac{\Gamma(k-1)}{(4\pi m)^{k-1}} \sum_{f \in \mathcal{F}} \overline{a_f(m)} f
\end{aligned} \tag{128}$$

where Theorem 1.7.3 gives the last equality. Continuing:

$$\begin{aligned}
P_{m,k}(z) &= \frac{\Gamma(k-1)}{(4\pi m)^{k-1}} \sum_{f \in \mathcal{F}} \overline{a_f(m)} \sum_{n \geq 1} a_f(n) e(nz) \\
&= \sum_{n \geq 1} \left(\frac{\Gamma(k-1)}{(4\pi m)^{k-1}} \sum_{f \in \mathcal{F}} a_f(n) \overline{a_f(m)} \right) e(nz)
\end{aligned} \tag{129}$$

Thus $a_{P_{m,k}}(n) = \sum_{f \in \mathcal{F}} a_f(n) \overline{a_f(m)}$. By Theorem 1.6.2,

$$\frac{\Gamma(k-1)}{(4\pi m)^{k-1}} \sum_{f \in \mathcal{F}} a_f(n) \overline{a_f(m)} = \delta(m, n) + \frac{2\pi}{i^k} \sum_{c \geq 1} \frac{1}{c} S(m, n; c) J_{k-1} \left(\frac{4\pi \sqrt{mn}}{c} \right) \tag{130}$$

□

2. L Functions and Hecke Operators

2.1. L Functions and Hecke Λ Function.

Definition 2.1.1

$$\Gamma(q) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) \mid \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{q} \right\} \leq \mathrm{SL}_2(\mathbb{Z}) \quad (131)$$

has index equal to $|\mathrm{SL}_2(\mathbb{Z}_q)|$ and

$$\Gamma_0(q) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) \mid q \mid c \right\} \quad (132)$$

. Note $\mathrm{SL}_2(\mathbb{Z}) = \Gamma_0(1)$.

Definition 2.1.2 $M_k(q, \chi)$, where $q \in \mathbb{N}$ and $\chi : \Gamma_0(q) \rightarrow \mathbb{C}$, is the space of functions $f : \mathbb{H} \rightarrow \mathbb{C}$ such that $\forall g \in \Gamma_0(q)$

$$f(gz) = \chi(g)j_k(g, z)f(z) \quad (133)$$

$S_k(q, \chi)$ is the corresponding space of cusp forms. For brevity, $S_k = S_k(1, 1)$ and $S_k(q) = S_k(q, 1)$.

Until otherwise specified, let $\tilde{\chi}$ be Dirichlet character mod q . Then $\chi : \Gamma_0(q) \rightarrow \mathbb{C}$ is defined as follows $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \rightarrow \tilde{\chi}(d)$.

Definition 2.1.3 For $s \in \mathbb{C}$ such that integral makes sense, define **Hecke Λ -Function**

$$\Lambda(f, s) := \int_0^\infty f(iy)y^s \frac{dy}{y} \quad (134)$$

Proposition 2.1.4 For $f \in S_k$, let $(a_n)_{n \geq 1}$ be its Fourier expansion at ∞ . Then for $\Re(s) > k + 1$

$$\Lambda(f, s) := (2\pi)^{-s} \Gamma(s) \sum_{n \geq 1} a_n n^{-s} \quad (135)$$

Proof

$$\begin{aligned} \Lambda(f, s) &:= \int_0^\infty f(iy)y^s \frac{dy}{y} \\ &= \int_0^\infty \sum_{n \geq 1} a_n e^{-2\pi ny} y^s \frac{dy}{y} \end{aligned} \quad (136)$$

We have to justify the swap of integral and sum,

$$\begin{aligned}
\int_0^\infty \left| \sum_{n \geq 1} a_n e^{-2\pi n y} \right| y^{s-1} dy &= \int_0^\infty \sum_{n \geq 1} |a_n| e^{-2\pi n y} y^{\Re(s)-1} dy \\
&= \sum_{n \geq 1} \int_0^\infty |a_n| e^{-2\pi n y} y^{\Re(s)-1} dy \quad (\text{By Tonellis Theorem}) \\
&= \sum_{n \geq 1} |a_n| \int_0^\infty e^{-2\pi n y} y^{\Re(s)-1} dy \\
&= \sum_{n \geq 1} |a_n| \int_0^\infty e^{-2\pi n y} y^{\Re(s)-1} dy \\
&= \sum_{n \geq 1} \frac{|a_n|}{2\pi n} \int_0^\infty e^{-u} \left(\frac{u}{2\pi n} \right)^{\Re(s)-1} du \\
&= \sum_{n \geq 1} \frac{|a_n|}{(2\pi n)^{\Re(s)}} \int_0^\infty e^{-u} u^{\Re(s)-1} du \\
&= (-2\pi)^{\Re(s)} \Gamma(\Re(s)) \sum_{n \geq 1} |a_n| n^{-\Re(s)}
\end{aligned} \tag{137}$$

By Corollary 2.1.10, $\sum_{n \geq 1} |a_n| n^{-\Re(s)}$ converges if $\Re(s) > \frac{k}{2} + 1$. Thus by Fubini's theorem, the interchange of integral and sum is allowed if $\Re(s) > \frac{k}{2} + 1$.

$$\begin{aligned}
\Lambda(f, s) &= \sum_{n \geq 1} a_n \int_0^\infty e^{-2\pi n y} y^s \frac{dy}{y} \\
&= \sum_{n \geq 1} a_n \left(\frac{1}{2\pi n} \right) \int_0^\infty e^{-u} \left(\frac{u}{2\pi n} \right)^{s-1} du \\
&= (2\pi)^{-s} \left(\int_0^\infty e^{-u} u^{s-1} du \right) \sum_{n \geq 1} a_n n^{-s} \\
&= (2\pi)^{-s} \Gamma(s) \sum_{n \geq 1} a_n n^{-s}
\end{aligned} \tag{138}$$

□

Thus the Hecke Λ -Function is the Dirichlet generating series of the Fourier coefficients of $f \in S_k$.

Proposition 2.1.5 $\forall f \in S_k$,

$$\Lambda(f, s) = i^k \Lambda(f, k - s) \tag{139}$$

Proof Recall that,

$$f\left(-\frac{1}{z}\right) = z^k f(z) \Rightarrow f(z) = \frac{f\left(-\frac{1}{z}\right)}{z^k} \tag{140}$$

Furthermore note, that $-\frac{1}{iy} = \frac{i}{y}$. Thus,

$$\begin{aligned}
\Lambda(f, s) &= \int_0^\infty f(iy) y^s \frac{dy}{y} \\
&= \int_0^\infty f\left(-\frac{1}{iy}\right) (iy)^{-k} y^s \frac{dy}{y} \\
&= i^k \int_0^\infty f\left(\frac{i}{y}\right) y^{k-s} \frac{dy}{y}
\end{aligned} \tag{141}$$

Let $u = \frac{1}{y} \Rightarrow du = -\frac{1}{y^2} dy \Rightarrow -\frac{1}{u} du = -y du = \frac{1}{y} dy$. Thus,

$$\Lambda(f, s) = i^k \int_0^\infty f(iu) y^{k-s} \frac{du}{u} = i^k \Lambda(f, k-s) \tag{142}$$

□

Thus there is a functional equation between $\Lambda(f, s)$ and $\Lambda(f, k-s)$ for $f \in S_k$. We expect something similar for general f .

Definition 2.1.6 For s such that $\Re(s)$ large enough that the sum converges, one defines

$$L(f, s) := \sum_{n \geq 1} a_n n^{-s} \tag{143}$$

which is called the **Hecke L-Function** of f

Lemma 2.1.7 For $f \in S_k(q, \chi)$ then the function $z \rightarrow (\Im(z))^{\frac{k}{2}} |f(z)|$ is bounded on $\mathbb{H}$.

Proof For all cusps c , $|f(z)| \rightarrow 0$ as $z \rightarrow c$ exponentially fast. Furthermore $g(z) = (\Im(z))^{\frac{k}{2}} |f(z)|$. Then $\forall \gamma \in \Gamma_0(q)$,

$$\begin{aligned}
g(\gamma z) &= (\Im(gz))^{\frac{k}{2}} |f(\gamma z)| \\
&= \left(\Im(z) |j_1(g, z)|^2 \right)^{\frac{k}{2}} |j_{-k}(g, z) f(z)| \\
&= \Im(z) |f(z)|
\end{aligned} \tag{144}$$

Thus g is $\Gamma_0(q, \chi)$. □

Lemma 2.1.8 Let $f \in S_k(q, \chi)$ and $(a_n)_{n \geq 0}$ is its fourier coefficients at ∞ . We have

$$\sum_{n \leq X} |a_n|^2 \ll X^k \tag{145}$$

where the implied constant depends on f .

Proof

$$\begin{aligned}
f(x + iy) &= \sum_{n \geq 0} a_n e(n(x + iy)) \\
&= a_0 + \sum_{n \geq 1} a_n e(nx) e^{-2\pi y}
\end{aligned} \tag{146}$$

Thus

$$|f(x + iy)|^2 \leq |a_0|^2 + \sum_{n \geq 1} |a_n|^2 e^{-4\pi y} \quad (147)$$

Thus $\forall y \geq 0$,

$$\int_0^1 |f(x + iy)|^2 dx \leq \int_0^1 \left[|a_0|^2 + \sum_{n \geq 1} |a_n|^2 e^{-4\pi y} \right] dx = |a_0|^2 + \sum_{n \geq 1} |a_n|^2 e^{-4\pi y} < (148)$$

Thus $f(x + iy) \in L^2(S^1)$. By Parseval's Identity,

$$\sum_{n \geq 0} |a_n|^2 e^{-4\pi n y} = \int_0^1 |f(x + iy)|^2 dx \quad (149)$$

Let $y = \frac{1}{X}$.

$$\sum_{0 \leq n \leq X} |a_n|^2 e^{-4\pi n/X} \leq \sum_{n \geq 0} |a_n|^2 e^{-4\pi n/X} = \int_0^1 \left| f\left(t + \frac{i}{X}\right) \right|^2 dt \quad (150)$$

Note $e^{-4\pi} \leq e^{-4\pi n X}$ for all $n \in \{0, \dots, X\}$. Thus

$$\sum_{0 \leq n \leq X} |a_n|^2 e^{-4\pi} \leq \int_0^1 \left| f\left(t + \frac{i}{X}\right) \right|^2 dt \quad (151)$$

Thus,

$$\sum_{0 \leq n \leq X} |a_n|^2 \leq e^{4\pi} \int_0^1 \left| f\left(t + \frac{i}{X}\right) \right|^2 dt \quad (152)$$

By Lemma 2.1.7, $\exists C_f$ such that $\mathfrak{I}(z)^{\frac{k}{2}} |f(z)| \leq C_f \Rightarrow |f(z)| \leq C_f \mathfrak{I}(z)^{-\frac{k}{2}}$.

Thus,

$$\sum_{n=0}^X |a_n|^2 \leq e^{4\pi} \int_0^1 C_f \mathfrak{I}\left(t + \frac{i}{X}\right)^{-k} dt = e^{4\pi} C_f X^k \quad (153)$$

Thus $\sum_{n=0}^X |a_n|^2 \ll X^k$ □

Theorem 2.1.9 Let $f \in S_k(q, \chi)$ and let $f(z) = \sum_{n \geq 1} a_n e(nz)$ be its Fourier expansion at ∞ . Then $(a_n)_{n \geq 0}$ has polynomial growth.

Proof $\forall y \in (0, \infty)$,

$$\begin{aligned} a_n &= \int_0^1 f(x + iy) e(n(x + iy)) dx \\ &= e^{2\pi n y} \int_0^1 f(x + iy) e(nx) dx \end{aligned} \quad (154)$$

Thus,

$$|a_n| \leq e^{2\pi ny} \int_0^1 |f(x + iy)| dx \quad (155)$$

By Lemma 2.1.7, $\exists C_f \ni \mathfrak{I}(z)^{\frac{k}{2}} |f(z)| \leq C_f$. Thus

$$\begin{aligned} |a_n| &\leq e^{2\pi ny} \int_0^1 C_f \mathfrak{I}(x + iy)^{-\frac{k}{2}} dy \\ &= C_f e^{2\pi ny} y^{-\frac{k}{2}} \end{aligned} \quad (156)$$

Take $y = \frac{1}{n}$. Then

$$|a_n| \leq C_f e^{2\pi} n^{\frac{k}{2}} \quad (157)$$

Thus $|a_n| = O\left(n^{\frac{k}{2}}\right)$.

□

Corollary 2.1.10 The Hecke L-Function is holomorphic for $\Re(s) > \frac{k}{2} + 1$

Proof Recall the Hecke L Function is given by:

$$L(f, s) := \sum_{n \geq 1} a_n n^{-s}. \quad (158)$$

By Theorem 2.1.9, $\exists C_f \ni |a_n| \leq C_f n^{\frac{k}{2}}$ thus

$$\sum_{n \geq 1} |a_n| n^{-s} \leq \sum_{n \geq 1} C_f n^{\frac{k}{2}-s} = C_f \sum_{n=1}^{\infty} n^{\frac{k}{2}-s} \quad (159)$$

which converges if $\frac{k}{2} - \Re(s) < -1 \Rightarrow \Re(s) > \frac{k}{2} + 1$. Thus $L(f, s)$ converges absolutely for $\Re(s) > \frac{k}{2} + 1$ so it converges locally uniformly (by the Weierstrass M Test). As $L(f, s)$ is a series of holomorphic functions that locally uniformly converges, it is a holomorphic function.

□

Lemma 2.1.11

$$\begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix}^{-1} \Gamma_0(q) \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} = \Gamma_0(q) \quad (160)$$

and χ is mapped to $\bar{\chi}$ by

$$\bar{\chi}(g) = \chi \left(\begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix}^{-1} g \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} \right) \quad (161)$$

Proof Let $\begin{pmatrix} a & b \\ qc & d \end{pmatrix} \in \Gamma_0(q)$ where

$$\begin{aligned}
\begin{pmatrix} 0 & \frac{1}{q} \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ qc & d \end{pmatrix} \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} &= \begin{pmatrix} 0 & \frac{1}{q} \\ -1 & 0 \end{pmatrix} \begin{pmatrix} bq & -a \\ qd & -qc \end{pmatrix} \\
&= \begin{pmatrix} d & -c \\ -bq & a \end{pmatrix} \in \Gamma_0(q)
\end{aligned} \tag{162}$$

$$\begin{aligned}
\chi \left(\begin{pmatrix} 0 & \frac{1}{q} \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ qc & d \end{pmatrix} \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} \right) &= \chi \left(\begin{pmatrix} d & -c \\ -bq & a \end{pmatrix} \right) \\
&= \chi(a) = \chi(a^{-1}) = \bar{\chi}(d) = \bar{\chi} \left(\begin{pmatrix} a & b \\ qc & d \end{pmatrix} \right)
\end{aligned} \tag{163}$$

□

Definition 2.1.12 Let $f \in S_k$, $\forall g \in \text{GL}_2(\mathbb{Q})$ then $(f|_g)(z) = (\det(g))^{\frac{k}{2}} j_k(g, z) f(gz)$

Theorem 2.1.13 (Hecke's Theorem) Let $f \in S_k(q, \chi)$. Then $\Lambda(f, s)$ and $L(f, s)$ have analytic continuation to $\mathbb{C}$. More precisely the function

$$\tilde{f}(z) = \left(f|_k \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} \right)(z) \tag{164}$$

is in $S(q, \bar{\chi})$ and for all $s \in \mathbb{C}$, we have

$$q^{\frac{1}{2}} \Lambda(f, s) = i^k \Lambda(\tilde{f}, k-s) q^{\frac{k-s}{2}} \tag{165}$$

Proof $\forall g \in \Gamma_0(q)$,

$$\begin{aligned}
\tilde{f}|_k g &= f|_k \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} |_k g \\
&= f|_k \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} g
\end{aligned} \tag{166}$$

By Lemma 2.1.11, $\exists g' \in \Gamma_0(q)$ such that $\begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} g = g' \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix}$. Thus

$$\begin{aligned}
f|_k g' \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} &= f|_k g' |_k \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} \\
&= \bar{\chi}(d) f|_k \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} \\
&= \bar{\chi}(d) \tilde{f}
\end{aligned} \tag{167}$$

Thus $f \in \Gamma_0(q, \bar{\chi})$. To show analytic continuation, we will start with the definition of $\Lambda(f, s)$.

$$\Lambda(f, s) = \int_0^\infty f(iy) y^s \frac{dy}{y}. \tag{168}$$

Lets split the integral $\frac{1}{\sqrt{q}}$. Thus

$$\Lambda(f, s) = \int_0^{\frac{1}{\sqrt{q}}} f(iy)y^s \frac{dy}{y} + \int_{\frac{1}{\sqrt{q}}}^{\infty} f(iy)y^s \frac{dy}{y}. \quad (169)$$

Lets start the first part of the integral. Let $u = \frac{1}{qy}$, then $du = -\frac{1}{qy^2} dy$. Thus $\frac{dy}{y} = -qy du = -\frac{du}{u}$. Thus

$$\begin{aligned} \int_0^{\frac{1}{\sqrt{q}}} f(iy)y^s \frac{dy}{y} &= - \int_{\infty}^{\frac{1}{\sqrt{q}}} f\left(\frac{i}{qu}\right)(qu)^{-s} \frac{du}{u} \\ &= \int_{\frac{1}{\sqrt{q}}}^{\infty} f\left(\frac{i}{qu}\right)(qu)^{-s} \frac{du}{u} \end{aligned} \quad (170)$$

Lets use Definition 2.1.12 (Modularity for $\mathrm{GL}_2(\mathbb{Z})$):

$$\begin{aligned} f\left(\frac{i}{qu}\right) &= f\left(-\frac{1}{qui}\right) \\ &= f\left(\begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} \cdot iu\right) \\ &= q^{-\frac{k}{2}}(qui)^k \tilde{f}(iu) \\ &= q^{\frac{k}{2}}(iu)^k \tilde{f}(iu) \end{aligned} \quad (171)$$

Thus,

$$\begin{aligned} \int_0^{\frac{1}{\sqrt{q}}} f(iy)y^s \frac{dy}{y} &= \int_{\frac{1}{\sqrt{q}}}^{\infty} q^{\frac{k}{2}}(iu)^k \tilde{f}(iu)(qu)^{-s} \frac{du}{u} \\ &= i^k q^{\frac{k}{2}-s} \int_{\frac{1}{\sqrt{q}}}^{\infty} \tilde{f}(iu)u^{k-s} \frac{du}{u}. \end{aligned} \quad (172)$$

Thus,

$$\Lambda(f, s) = \int_{\frac{1}{\sqrt{q}}}^{\infty} \left[i^k q^{\frac{k}{2}-s} \tilde{f}(iu)y^{k-s} + f(iy)y^s \right] \frac{dy}{y}. \quad (173)$$

Let's do the same thing with $\Lambda(\tilde{f}, s)$ (we can do this because $\tilde{f}$ is also a cusp form, and the integral exists for all $s \in \mathbb{C}$ and defines a entire function). Thus

$$\Lambda(\tilde{f}, s) = \int_{\frac{1}{\sqrt{q}}}^{\infty} \left[i^k q^{\frac{k}{2}-s} \tilde{\tilde{f}}(iy)y^{k-s} + \tilde{f}(iy)y^s \right] \frac{dy}{y}. \quad (174)$$

We want to get some equality between $\Lambda(f, s)$ and $\Lambda(\tilde{f}, s)$. Lets use the relation between, f and $\tilde{f}$.

$$\tilde{\tilde{f}}(z) = \left(f|_k \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix}^2 \right)(z) = \left(f|_q \begin{pmatrix} -q & 0 \\ 0 & -q \end{pmatrix} \right) = q^k(-q)^{-k} f(z) = (-1)^k f(z) \quad (175)$$

Thus

$$\Lambda(\tilde{f}, s) = \int_{\frac{1}{\sqrt{q}}}^{\infty} [(-i)^k q^{\frac{k}{2}-s} f(iy) y^{k-s} + \tilde{f}(iy) y^s] \frac{dy}{y}. \quad (176)$$

After some algebra we see, $q^{\frac{1}{2}} \Lambda(f, s) = i^k q^{\frac{k-s}{2}} \Lambda(\tilde{f}, k-s)$

□

The relationship between $s \leftrightarrow k-s$ suggests that $\Re(s) = \frac{k}{2}$ has critical behavior. We call the line the “critical line”. Quite often we have $\tilde{f} = f$.

Example 2.1.14 Although E_k isn't a cusp form, we can define an analogous Hecke Λ function and L function in the same way. Where $L(E_k, s) := \sum_{n \geq 1} a_n n^{-s}$, thus you skip a_0 . Up to a constant multiple $a_n \approx \sigma_{k-1}(n)$. Thus

$$\begin{aligned} L(E_k, s) &\approx \sum_{n \geq 1} \sigma_{k-1}(n) n^{-s} \\ &= \sum_{n \geq 1} \sum_{d|n} d^{k-1} n^{-s} \\ &= \left(\sum_{n \geq 1} \frac{1}{n^{-s}} \right) \left(\sum_{n \geq 1} \frac{1}{n^{s-k-1}} \right) \\ &= \zeta(s) \zeta(s-k-1) \end{aligned} \quad (177)$$

Thus the function has a Euler Product thus reflecting multiplicativity.

2.2. Hecke Operator

Lemma 2.2.1 Let $n \geq 1$ be a. Let $\Delta_n := \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \mid ad = n, 0 \leq b < d \right\}$. Then Δ_n parametrizes the $\mathrm{SL}_2(\mathbb{Z})$ orbits of the subset

$$G_n = \{g \in M_2(\mathbb{Z}) \mid \det(g) = n\} \quad (178)$$

for $\mathrm{SL}_2(\mathbb{Z})$ by left multiplication. There is a disjoint union

$$G_n = \bigsqcup_{g \in \Delta_n} \mathrm{SL}_2(\mathbb{Z})g \quad (179)$$

Proof Let $\gamma_1, \gamma_2 \in \mathrm{SL}_2(\mathbb{Z})$ and $g_1, g_2 \in \Gamma_n$. Suppose $\gamma_1 g_1 = \gamma_2 g_2$. Let $\gamma = \begin{pmatrix} q & r \\ s & t \end{pmatrix} = \gamma_2^{-1} \gamma_1 \in \mathrm{SL}_2(\mathbb{Z})$. $\gamma g_1 = g_2$. Thus

$$\begin{pmatrix} qa_1 & qb_1 + rd_1 \\ sa_1 & sb_1 + td_1 \end{pmatrix} = \begin{pmatrix} q & r \\ s & t \end{pmatrix} \begin{pmatrix} a_1 & b_1 \\ 0 & d_1 \end{pmatrix} = \begin{pmatrix} a_2 & b_2 \\ 0 & d_2 \end{pmatrix}. \quad (180)$$

Thus $a_2 = sa_1 \Rightarrow s = 0$ (contradiction if $a_1 = 0$). Thus $\begin{pmatrix} q & r \\ 0 & t \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) \Rightarrow q, t = \pm 1$. Thus $\pm d_1 = td_1 = sb_1 + td_1 = d_2$ and $\pm a_1 = qa_1 = a_2$. But by assumption $d_2 \geq 0$ and $a_2 \geq 0$. Thus $q, t = 1$. Thus

$$\begin{pmatrix} a_1 & b_1 + rd_1 \\ 0 & d_1 \end{pmatrix} = \begin{pmatrix} a_2 & b_2 \\ 0 & d_2 \end{pmatrix} \quad (181)$$

. Since $0 \leq b_1, b_2 < d_1 \Rightarrow r = 0$. Thus $\gamma = I$. Thus $\gamma_1 = \gamma_2$ and $g_1 = g_2$. Thus the union is disjoint.

$\forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G_n$. Let $\gamma = \frac{c}{(a,c)}$ and $\delta = -\frac{a}{(a,c)}$. $(\gamma, \delta) = 1 \Rightarrow \exists \alpha, \beta \in \mathbb{Z} \ni \alpha\delta + \beta\gamma = 1$. Thus $\begin{pmatrix} \alpha & -\beta \\ \gamma & \delta \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$.

$$\begin{aligned} \begin{pmatrix} \alpha & -\beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} &= \begin{pmatrix} \alpha a - \beta c & \alpha b - \beta d \\ 0 & \frac{cb - ad}{(a,c)} \end{pmatrix} \\ &= \begin{pmatrix} (a,c) & \alpha b - \beta d \\ 0 & \frac{n}{(a,c)} \end{pmatrix} = \begin{pmatrix} a_1 & b_1 \\ 0 & d_1 \end{pmatrix} \end{aligned} \quad (182)$$

where $a_1 d_1 = n$ and up to multiplying -1 we have that $a_1, d_1 \geq 1$. Exists $u \in \mathbb{Z}$ such that $b_1 + u d_1 = b_1 \bmod d_1$. Thus

$$\begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha & -\beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a_1 & b_1 \\ 0 & d_1 \end{pmatrix} = \begin{pmatrix} a_1 & b_1 \bmod d_1 \\ 0 & d_1 \end{pmatrix} \in \Delta_n \quad (183)$$

Thus $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z}) \Delta_n$ □

Definition 2.2.2 Fix $k, q \geq 1$, and a Dirichlet character $\chi \bmod q$. Let $P_k := \{f : \mathbb{H} \rightarrow \mathbb{C} \mid f \text{ is } 1\text{-periodic}\}$. Let $T_n : P_k \rightarrow P_k$ (this depends on q, k) by

$$(T_n f)(z) = n^{\frac{k}{2}-1} \sum_{g \in \Delta_n} \chi(a) (f|_k g)(z) \quad (184)$$

For general $f : \mathbb{H} \rightarrow \mathbb{C}$ 1 periodic,

$$(f|_k g)(z) := (\det(g))^{\frac{k}{2}} j_k(g, z) f(gz) \quad (185)$$

Thus

$$\begin{aligned} (T_n f)(z) &= n^{\frac{k}{2}-1} \sum_{g \in \Delta_n} \chi(a) n^{\frac{k}{2}} j_k(g, z) f(gz) \\ &= n^{-1} \sum_{\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \Delta_n} \chi(a) n^k d^{-k} f\left(\frac{az+b}{d}\right) \\ &= n^{-1} \sum_{ad=n} \chi(a) a^k \sum_{b=0}^{d-1} f\left(\frac{az+b}{d}\right) \end{aligned} \quad (186)$$

Lemma 2.2.3 For $g \in \Delta_n$, there is a unique $v \in \mathbb{Z}$ and $h \in \Delta_n$ such that

$$g \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & v \\ 0 & 1 \end{pmatrix} h \quad (187)$$

and moreover $g \rightarrow h$ is bijective and $\chi(a) = \chi(a')$

Proof $\forall \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \Delta_n$.

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & a+b \\ 0 & d \end{pmatrix} \quad (188)$$

$\exists! v \in \mathbb{Z} \ni a + b + vd = (a + b) \bmod d$. Thus

$$\begin{pmatrix} 1 & v \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & a+b \\ 0 & d \end{pmatrix} = \begin{pmatrix} a & (a+b) \bmod d \\ 0 & d \end{pmatrix} \in \Delta_n \quad (189)$$

Thus

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -v \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & (a+b) \bmod d \\ 0 & d \end{pmatrix}. \quad (190)$$

Let $f : \Delta_n \rightarrow \Delta_n$ where $\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \rightarrow \begin{pmatrix} a & (a+b) \bmod d \\ 0 & d \end{pmatrix}$. $\forall \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}, \begin{pmatrix} p & q \\ 0 & r \end{pmatrix} \in \Delta_n$ where

$$\begin{pmatrix} a & (a+b) \bmod d \\ 0 & d \end{pmatrix} = \begin{pmatrix} p & (p+q) \bmod r \\ 0 & r \end{pmatrix} \quad (191)$$

Thus $a = p$ and $d = r$. Since $+_a : \mathbb{Z}_d \rightarrow \mathbb{Z}_d$ where $x \rightarrow a + x$ is a bijection, $b = q$. Thus f is injective. Surjective, $\forall \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in \Delta_n$.

$$f\left(\begin{pmatrix} a & (b-a) \bmod d \\ 0 & d \end{pmatrix}\right) = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}. \quad (192)$$

□

Lemma 2.2.4 $T(n)$ is well defined for f is 1 periodic so is the function $T(n)f$ is defined above

Proof

$$\begin{aligned} T(n)(f)|_k \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} &= n^{\frac{k}{2}-1} \sum_{g \in \Delta_n} \chi(a)f|_k g|_k \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ &= n^{\frac{k}{2}-1} \sum_{h \in \Delta_n} \chi(a)f|_k \begin{pmatrix} 1 & v_g \\ 0 & 1 \end{pmatrix} |_k h \\ &= n^{\frac{k}{2}-1} \sum_{h \in \Delta_n} \chi(a)f|_k h \\ &= T(n)(f) \end{aligned} \quad (193)$$

The 3rd equality comes from the bijection between $g \leftrightarrow h$ and 1-periodicity. □

Corollary 2.2.5 Suppose $f \in P_k$ satisfies

$$f(z) = \sum_{m \geq 0} a_m e(mz) \quad (194)$$

Then

$$T(n)f(z) = \sum_{m \geq 0} \left(\sum_{d|(m,n)} \chi(d) d^{k-1} a\left(\frac{mn}{d^2}\right) \right) e(mz) \quad (195)$$

There is a surprising symmetry between m and n .

Proof By Definition,

$$\begin{aligned} (T_n f)(z) &= n^{\frac{k}{2}-1} \sum_{g \in \Delta_n} (f|_k g)(z) \\ &= n^{-1} \sum_{ad=n} \chi(a) a^k \sum_{b=0}^{d-1} f\left(\frac{az+b}{d}\right) \\ &= n^{-1} \sum_{ad=n} \chi(a) a^k \sum_{b=0}^{d-1} \sum_{m=0}^{\infty} a_m e\left(\frac{maz+mb}{d}\right) \\ &= n^{-1} \sum_{m=0}^{\infty} a_m \sum_{ad=n} \chi(a) a^k \sum_{b=0}^{d-1} e\left(\frac{maz+mb}{d}\right) \\ &= n^{-1} \sum_{m=0}^{\infty} a_m \sum_{ad=n} \chi(a) a^k e\left(\frac{maz}{d}\right) \sum_{b=0}^{d-1} e\left(\frac{mb}{d}\right) \\ &= n^{-1} \sum_{m=0}^{\infty} a_m \sum_{ad=n} \chi(a) a^k e\left(\frac{maz}{d}\right) \sum_{b=0}^{d-1} e\left(\frac{mb}{d}\right) \end{aligned} \quad (196)$$

Since

$$\sum_{b=0}^{d-1} e\left(\frac{mb}{d}\right) = \begin{cases} d & \text{if } d \mid m \\ 0 & \text{otherwise} \end{cases}, \quad (197)$$

$$\begin{aligned} (T(n)f)(z) &= \sum_{m=0}^{\infty} a_m \sum_{\substack{ad=n \\ d|m}} \chi(a) \frac{d}{n} a^k e\left(\frac{maz}{d}\right) \\ &= \sum_{m=0}^{\infty} a_m \sum_{\substack{ad=n \\ d|m}} \chi(a) a^{k-1} e\left(\frac{maz}{d}\right) \\ &= \sum_{\substack{ad=n \\ a,d \geq 1}} \chi(a) a^{k-1} \sum_{l=0}^{\infty} a_{ld} e(laz) \\ &= \sum_{r=0}^{\infty} e(rz) \sum_{\substack{ad=n \\ a|m}} \chi(a) a^{k-1} a_{\frac{md}{a}} \\ &= \sum_{r=0}^{\infty} e(rz) \sum_{\substack{ad=n \\ a|r}} \chi(a) a^{k-1} a_{\frac{mn}{a^2}} \\ &= \sum_{r=0}^{\infty} e(rz) \sum_{a|(r,n)} \chi(a) a^{k-1} a_{\frac{mn}{a^2}} \end{aligned} \quad (198)$$

□

Corollary 2.2.6 For all $f \in P_k$ with fourier expansion $f(z) = \sum_{m=0}^{\infty} a_f(m)e(mz)$

$$\begin{aligned} a_{T_n(f)}(0) &= a_f(0) \sum_{d|n} \chi(a) a^{k-1} \\ a_{T_n(f)}(1) &= a_f(n) \end{aligned} \quad (199)$$

If $T_n(f) = \lambda f$ then $a_f(0) = 0$ or $\sum_{d|n} \chi(a) a^{k-1} = 0$. If $T_n(f) = \lambda f$ then $a_f(n) = \lambda a_f(1)$.

Lemma 2.2.7 For $g \in \Delta_n$ and $\gamma \in \Gamma_0(d)$, $\exists!(h, \eta) \in \Delta_n, \Gamma_0(d)$ such that

$$g\gamma = \eta h \quad (200)$$

The map $g \rightarrow h$ is a bijection.

Proof Suppose

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \gamma = \gamma' \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix}. \quad (201)$$

Let $\gamma = \begin{pmatrix} r & s \\ t & u \end{pmatrix}$ and $\gamma' = \begin{pmatrix} r' & s' \\ t' & u' \end{pmatrix}$.

$$\begin{pmatrix} ar + bt & as + bu \\ dt & du \end{pmatrix} = \begin{pmatrix} a'r' & r'b' + s'd' \\ t'a' & t'b' + u'd' \end{pmatrix} \quad (202)$$

Since $q \mid t, t', ar \equiv a'r' \pmod{q}$. Note $(r, q) = (r', q) = 1$. If $(a, q) = 1, (a', q) = 1$. $dt = t'a' \equiv 0 \pmod{q}$. Since $(a', q) = 1, t' = 0 \pmod{q}$. thus $\gamma' \in \Gamma_0(q)$. □

Theorem 2.2.8 The Hecke operators give linear maps

$$\begin{aligned} M_k(q, \chi) &\rightarrow M_k(q, \chi) \\ S_k(q, \chi) &\rightarrow S_k(q, \chi) \end{aligned} \quad (203)$$

Proof Follows periodicity proof closely. Let $\gamma \in \Gamma_0(q)$,

$$\begin{aligned} (T_n f |_k \gamma &= n^{\frac{k}{2}-1} \sum_{g \in \Delta_n} \chi(a) f |_k g |_k \gamma \\ &= n^{\frac{k}{2}-1} \sum_{g \in \Delta_n} \chi(a) \chi(\eta) f |_k \eta |_k h \\ &= \chi(\gamma) n^{\frac{k}{2}-1} \sum_{h \in \Delta_n} \chi(a') f |_k h \\ &= \chi(\gamma) (T_n f) \end{aligned} \quad (204)$$

Thus T_n is a map from $M_k(q, \chi)$ to itself and is linear. Furthermore by Corollary 2.2.6, $S_k(q, \chi) \rightarrow S_k(q, \chi)$

□

Theorem 2.2.9 The Hecke operator on $M_k(q, \chi)$ follows the relations

$$T(m)T(n) = \sum_{d|(m,n)} \chi(d) d^{k-1} T\left(\frac{mn}{d^2}\right) \quad (205)$$

In particular,

$$T(m)T(n) = T(mn) = T(n)T(m) \quad (206)$$

if $(m, n) = 1$.

Proof By definition

$$\begin{aligned} T(m)T(n)f &= n^{\frac{k}{2}-1} \sum_{a_1 c_1 = m} \sum_{b_1 = 0}^{c_1-1} \chi(a_1) \left(T(n)f|_k \begin{pmatrix} a_1 & b_1 \\ 0 & c_1 \end{pmatrix} \right) \\ &= (mn)^{\frac{k}{2}-1} \sum_{\substack{a_1 c_1 = m \\ a_2 c_2 = n}} \sum_{b_1 = 0}^{c_1-1} \sum_{b_2 = 0}^{c_2-1} \chi(a_1) \chi(a_2) \left(f|_k \begin{pmatrix} a_2 & b_2 \\ 0 & c_2 \end{pmatrix} |_k \begin{pmatrix} a_1 & b_1 \\ 0 & c_1 \end{pmatrix} \right) \\ &= (mn)^{\frac{k}{2}-1} \sum_{\substack{a_1 c_1 = m \\ a_2 c_2 = n}} \sum_{b_1 = 0}^{c_1-1} \sum_{b_2 = 0}^{c_2-1} \chi(a_1) \chi(a_2) \left(f|_k \begin{pmatrix} a_2 & b_2 \\ 0 & c_2 \end{pmatrix} \begin{pmatrix} a_1 & b_1 \\ 0 & c_1 \end{pmatrix} \right) \quad (207) \\ &= (mn)^{\frac{k}{2}-1} \sum_{\substack{a_1 c_1 = m \\ a_2 c_2 = n}} \sum_{b_1 = 0}^{c_1-1} \sum_{b_2 = 0}^{c_2-1} \chi(a_1) \chi(a_2) \left(f|_k \begin{pmatrix} a_1 a_2 & a_2 b_1 + b_2 c_1 \\ 0 & c_1 c_2 \end{pmatrix} \right) \\ &= (mn)^{\frac{k}{2}-1} \sum_{\substack{a_1 c_1 = m \\ a_2 c_2 = n}} \sum_{b_1 = 0}^{c_1-1} \sum_{b_2 = 0}^{c_2-1} \chi(a_1 a_2) \left(f|_k \begin{pmatrix} a_1 a_2 & a_2 b_1 + b_2 c_1 \\ 0 & c_1 c_2 \end{pmatrix} \right) \end{aligned}$$

If $(m, n) = 1$, then as $b_1, b_2, a_2 b_1 + b_2 c_1$ sums uniquely over (representatives of) $\mathbb{Z}_{d_1 d_2}$. Because of periodicity, the choice of representative doesn't matter. Thus if $(m, n) = 1$,

$$T(m)T(n)f = (mn)^{\frac{k}{2}-1} \sum_{ac=mn} \chi(a) \sum_{b=0}^{c-1} f|_k \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = T(mn)f \quad (208)$$

For the more general case we need a bit more work. $a_2 b_1 + b_2 c_1$ with a_2 and c_1 varying is always a multiple of $\delta = (a_1, c_2)$. Note $\delta \mid a_1 a_2$ and $\delta \mid c_1 c_2$. Thus we can split the sum according to δ and replace a_1 by δa_1 and c_2 by δc_2 . Thus

$$T(m)T(n)f = (mn)^{\frac{k}{2}-1} \sum_{\delta|(m,n)} \chi(\delta) \sum_{\substack{a_1 c_1 = \frac{m}{\delta} \\ a_2 c_2 = \frac{n}{\delta} \\ (a_1, c_2) = 1}} \sum_{b=0}^{c_1 c_2 - 1} \chi(a_1 a_2) \left(f|_k \begin{pmatrix} a_1 a_2 & b \\ 0 & c_1 c_2 \end{pmatrix} \right) \quad (209)$$

□

2.3. Simultaneous Diagonalization

For $n \geq 1$, $T(n) : S_{12}(1) \rightarrow S_{12}(1)$ is a linear map. Since $\dim S_k(1) = 1$, generated by $\Delta -$ function where

$$\begin{aligned}\Delta(z) &= (E_4^3 - E_6^2)(z) = e(z) \prod_{n \geq 1} (1 - e(nz))^{24} \\ &= \sum_{n \geq 1} \tau(n) e(nz)\end{aligned}\tag{210}$$

where $\tau(n)$ is the Ramanujan tau function. Since $T(n)\Delta = \lambda_n \Delta$ (dimensionality argument), then

$$\tau(n) = a_{T(n)\Delta}(1) = a_{\lambda_n \Delta}(1) = \lambda_n \tau(1).\tag{211}$$

We also get:

1. $\tau(m)\tau(n) = \tau(mn) \Leftrightarrow (m, n) = 1$ by the multiplicativity of hecke operators.
2. More generally

$$\tau(m)\tau(n) = \sum_{d|(m,n)} d^{11} \tau\left(\frac{mn}{d^2}\right)\tag{212}$$

3. Thus $\tau(p)\tau(p^v) = \tau(p^{v+1}) + p^{11} \tau\left(\frac{p^{v+1}}{p^2}\right) = \tau(p^{v+1}) + p^{11} \tau(p^{v-1}) \Rightarrow \tau(p^{v+1}) = \tau(p)\tau(p^v) - p^{11} \tau(p^{v-1})$

$$\begin{aligned}(1 - \tau(p)p^{-s} + p^{11}p^{-2s})(\tau(1) + \tau(p)p^{-s} + \tau(p^2)p^{-2s} + \dots) &= 1 \\ \Rightarrow \tau(1) + \tau(p)p^{-s} + \tau(p^2)p^{-2s} + \dots &= \frac{1}{1 - \tau(p)p + p^{11-2s}}\end{aligned}\tag{213}$$

Thus we obtain,

$$L(\Delta, s) = \sum_{n \geq 1} \frac{\tau(n)}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - \tau(p)p + p^{11-2s}}\tag{214}$$

for $\Re(s)$ large enough. We also obtain a analytic continuation to a entire function with

$$(2\pi)^{-s} \Gamma(s) L(\Delta, s) = (2\pi)^{s-12} \Delta(12-s) L(\Delta, 12-s)\tag{215}$$

This allows us to understand the Ramanujan conjecture that $|\tau(n)| \leq d(n)n^{\frac{11}{2}}$. Write $1 - \tau(p)x + p^{11}x^2 = (1 - \alpha_p x)(1 - \beta_p x)$. Then $\tau(p) = \alpha_p + \beta_p$ and $\alpha_p \beta_p = p^{11}$.

Theorem 2.3.1 (Ramanujan and Deligne) We have $|\alpha_p| = |\beta_p| = p^{\frac{11}{2}}$

Corollary 2.3.2 (Ramanujan and Deligne) We have $|\tau(n)| \leq d(n)n^{\frac{11}{2}}$

Proof Base case: $k = 1$, $\tau(p^k) = \alpha_p + \beta_p$

Inductive Hypothesis: Suppose $k \in \mathbb{N}$ that for all $n \leq k$, $\tau(p^n) = \alpha_p^k + \alpha_p^{k-1}\beta_p + \dots + \alpha_p\beta_p^{k-1} + \beta_p^k$.

Inductive step:

$$\begin{aligned}
\tau(p^{k+1}) &= \tau(p)\tau(p^k) - p^{11}\tau(p^{k-1}) \\
&= (\alpha_p + \beta_p)(\alpha_p^k + \alpha_p^{k-1}\beta_p + \dots + \alpha_p\beta_p^{k-1} + \beta_p^k) \\
&\quad - \alpha_p\beta_p(\alpha_p^{k-1} + \alpha_p^{k-2}\beta_p + \dots + \alpha_p\beta_p^{k-2} + \beta_p^{k-1}) \\
&= (\alpha_p^{k+1} + \alpha_p^k\beta_p + \dots + \alpha_p^2\beta_p^{k-1} + \alpha\beta_p^k) + (\alpha_p^k\beta_p + \alpha_p^{k-1}\beta_p^2 + \dots + \alpha_p\beta_p^k + \beta_p^{k+1}) \\
&\quad - (\alpha_p^k\beta_p + \alpha_p^{k-1}\beta_p^2 + \dots + \alpha_p^2\beta_p^{k-1} + \alpha_p\beta_p^k) \\
&= \alpha_p^{k+1} + \alpha_p^k\beta_p + \dots + \alpha_p\beta_p^k + \beta_p^{k+1}
\end{aligned}$$

Thus $\tau(p^n) = \sum_{m=0}^n \alpha_p^{n-m}\beta_p^m$. Thus $|\tau(p^n)| \leq \sum_{m=0}^n (p^{\frac{11}{2}})^n = (n+1)p^{\frac{11n}{2}} = d(p^n)p^{\frac{11n}{2}}$.
Thus $|\tau(n)| \leq d(n)n^{\frac{11}{2}}$ $\square$

Does this situation extend to other spaces of modular forms of dimension $k \geq 2$?

Lemma 2.3.3 Let E be a (finite dimensional) Hilbert Space and $(u_i)_{i \in I}$ a family of commuting *normal* operators. Then there exists a orthonormal basis of E such that all u_i are diagonal in this basis.

Proof Linear algebra. But quick summary:

1. Every normal $u \in \text{End}(E)$ is diagonalizable in orthonormal basis.
2. if u and v commute then v is a endomorphism of every eigenspace of u .

$\square$

Lemma 2.3.4 For $g = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \Delta_n$ we have

$$\int_D (f_1|_k g)(x+iy) \overline{(f_2|_k g')(x+iy)} y^k \frac{dx dy}{y^2} = \int_D f_1(x+iy) \overline{(f_2|_k g')(x+iy)} y^k \frac{dx dy}{y^2} \quad (217)$$

where $g' = \begin{pmatrix} d & -b \\ 0 & a \end{pmatrix}$

Proof $\forall \gamma \in \begin{pmatrix} p & qn \\ rn & s \end{pmatrix} \in \Gamma(n)$ where $\gamma \equiv I \pmod{n}$.

$$\begin{aligned}
g\gamma g^{-1} &= \frac{1}{n} \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \begin{pmatrix} p & qn \\ rn & s \end{pmatrix} \begin{pmatrix} d & -b \\ 0 & a \end{pmatrix} \\
&= \frac{1}{n} \begin{pmatrix} pn + bdnr & -abp + a^2nq - b^2nr + abs \\ d^2nr & ads - bdnr \end{pmatrix} \\
&= \frac{1}{n} \begin{pmatrix} pn + bdnr & -abp + a^2nq - b^2nr + abs \\ d^2nr & ns - bdnr \end{pmatrix}
\end{aligned} \quad (218)$$

It suffices to show that $\det(g\gamma g^{-1}) = 1$, which is true since $\det$ is a homomorphism, and the $g\gamma g^{-1}$ has entiger entries.

1. Top left: $\frac{pn+bdnr}{n} = p + bdr$
2. Top Right: $ab(s-p) + n(a^2q - b^2p) \equiv ab(s-p) \pmod{n}$. Note $s \equiv p \equiv 1 \pmod{n}$. Thus $s-p \equiv 0 \pmod{n}$. Thus the top right is a integer
3. Bottom row: trivially integers.

Thus $\exists g' \in \text{SL}_2(\mathbb{Z})$ where $g\gamma g^{-1} = g'$.

Thus

$$\int_D (f_1|_k g)(z) \overline{f_2(z)} y^k \frac{dx dy}{y^2} = n^{\frac{k}{2}} d^{-k} \int_D f_1(gz) \overline{f_2(z)} y^k \frac{dx dy}{y^2} \quad (219)$$

$w = gz = \frac{az+b}{d} = \frac{ax+b}{d} + \frac{aiy}{d} = u + iv$. Thus $y^{k-2} dx dy = \left(\frac{d}{a}\right)^k v^{k-2} du dv$

$$\begin{aligned} \int_D (f_1|_k g)(z) \overline{f_2(z)} y^k \frac{dx dy}{y^2} &= n^{\frac{k}{2}} d^{-k} \int_{g(D)} f_1(w) \overline{f_2(g'z)} \left(\frac{d}{a}\right)^k v^{k-2} du dv \\ &= n^{\frac{k}{2}} \int_{g(D)} f_1(w) \overline{f_2(g'z)} \left(\frac{1}{a}\right)^k v^{k-2} du dv \end{aligned} \quad (220)$$

$$(f_2|_k g')(z) = n^{\frac{k}{2}} a^{-k} f_2(g'z) \Rightarrow f_2(g'z) = n^{-\frac{k}{2}} a^k (f_2|_k g')(z).$$

Thus

$$\begin{aligned} \int_D (f_1|_k g)(z) \overline{f_2(z)} y^k \frac{dx dy}{y^2} &= n^{\frac{k}{2}} a^{-k} \int_D f_1(z) \overline{n^{-\frac{k}{2}} a^k (f_2|_k g')(z)} y^k \frac{dx dy}{y^2} \\ &= \int_D f_1(z) \overline{(f_2|_k g')(z)} y^k \frac{dx dy}{y^2} \end{aligned} \quad (221)$$

□

Lemma 2.3.5 Each $T(n)$ for $n \geq 1$ defines a normal endomorphism of $S_k(1)$ with respect to the Petersson inner product ie

$$\langle f, g \rangle = \int_D f(z) \overline{g(z)} y^k \frac{dx dy}{y} \quad (222)$$

Proof We will show that $T(p)^\dagger = T(p)$ which will imply self adjointness for all $T(n)$ by multiplicative.

$$\begin{aligned} \langle T(p)f_1, f_2 \rangle &= \int_D (T_p f_1)(z) \overline{f_2(z)} y^{k-2} dx dy \\ &= p^{\frac{k}{2}-1} \int_D \sum_{g \in \Delta_n} (f_1|_k g)(z) \overline{f_2(z)} y^{k-2} dx dy \\ &= p^{\frac{k}{2}-1} \sum_{g \in \Delta_p} \int_D (f_1|_k g)(z) \overline{f_2(z)} y^{k-2} dx dy \\ &= p^{\frac{k}{2}-1} \sum_{g \in \Delta_p} \int_D f_1(z) \overline{(f_2|_k g')(z)} y^{k-2} dx dy \end{aligned} \quad (223)$$

$g = \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}$ if $0 \leq b < p$ or $g = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$. Let $\gamma, \gamma' \in \text{SL}_2(\mathbb{Z})$ where

$$\gamma = \begin{cases} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} & \text{if } g = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \\ \begin{pmatrix} -b & -1 \\ 1 & 0 \end{pmatrix} & \text{if } g = \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix} \end{cases} \quad (224)$$

and

$$\gamma' = \begin{cases} \begin{pmatrix} p & -1 \\ 0 & 1 \end{pmatrix} & \text{if } g = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \\ \begin{pmatrix} 0 & -1 \\ 1 & b \end{pmatrix} & \text{if } g = \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix} \end{cases} \quad (225)$$

Note $\gamma' g' = g \gamma$.

Thus

$$\langle T_p f_1, f_2 \rangle = \langle f_1, T_p f_2 \rangle \quad (226)$$

Thus the hecke operator is self adjoint. $\square$

A corollary from this result is the fact that eigenvalues of $T(p)$ is real.

Theorem 2.3.6 (Hecke) For $k \geq 2$, there exists a *unique* basis $\mathcal{H}_k$ of $S_k(1)$ whose elements are eigenfunctions of all $T(n)$ for $n \geq 1$, and each element has a 1st fourier coefficients of 1.

Proof The previous 3 lemmas, give the existence of $\mathcal{H}_k$ the Hecke basis. Suppose f is a simultaneous eigenfunction of all $T(n)$ for $n \geq 1$. Then since $\lambda_n a_f(1) = a_{\lambda_n f}(1) = a_{T_n f}(1) = a_f(n)$. So $a_f(1) = 0 \Leftrightarrow f = 0$. Thus we can replace any $f \in \mathcal{H}_k$ with a scalar multiple to ensure $a_f(1) = 1$. Suppose f is a simultaneous eigenfunction of $T(n)$. Then $f = \sum_{i \in I} a_i f_i$. If $T(n) f_i = \lambda_{n,i} f_i$ and $T(n) f = \lambda_n f$. Thus we get

$$\sum_{i \in I} \lambda_n a_i f_i = \lambda_n f = T(n) f = \sum_{i \in I} a_i \lambda_{n,i} f_i \Rightarrow \lambda_n a_i = a_i \lambda_{n,i} \quad (227)$$

Take i such that $a_i \neq 0$, then we get $\lambda_n = \lambda_{n,i}$ for all n . But since $a_f(n) = \lambda_n a_f(1)$ we get $f = a_f(1) f_i$. $\square$

2.4. Distribution of Hecke Eigenvalues

Definition 2.4.1 Let X be a compact metric space. Let $(\mu_n)_{n \in \mathbb{N}}$ be a sequence of (Radon) measures on X and μ a Radon Measure on X . We say $\mu_n \rightarrow \mu$ (weakly in law) if $\forall f : X \rightarrow \mathbb{C}$ continuous,

$$\int_X f d\mu = \lim_{n \rightarrow \infty} \int f d\mu_n \quad (228)$$

We will apply this to X a compact set in $\mathbb{R}$, and measures μ_n of the type

$$\mu_n := \sum_{x \in X_n} \alpha_x \delta_x \quad (229)$$

where $X_n \subset X$ is a finite set. In particular, when $X_n \neq \emptyset$ and $\alpha_x = \frac{1}{|X_n|}$, then the resulting definition is of a μ -equidistribution

$$\int f d\mu = \lim_{n \rightarrow \infty} \frac{1}{|X_n|} \sum_{x \in X_n} f(x) \quad (230)$$

Definition 2.4.2 (*Chebyshev polynomials of the 2nd kind*) Let $(U_n : [-2, 2] \rightarrow \mathbb{C})_{n \in \mathbb{N}}$ be a collection of functions where $U_n(2 \cos \theta) = \frac{\sin((n+1)\theta)}{\sin(\theta)}$

Lemma 2.4.3 $(U_n)_{n \in \mathbb{N}}$ (Definition 2.4.2) form a orthonormal basis of $L^2([-2, 2], \mu_{\text{ST}})$ in particular $\int U_n \mu_{\text{ST}} = \int U_n U_0 \mu_{\text{ST}} = \begin{cases} 1 & \text{if } n=0 \\ 0 & \text{otherwise} \end{cases}$

Proof

$$\langle U_m, U_n \rangle = \int_{-2}^2 U_m(x) \overline{U_n(x)} d\mu_{\text{ST}} = \int_{-2}^2 U_m(x) U_n(x) d\mu_{\text{ST}} \quad (231)$$

Let $x = 2 \cos(\theta)$, thus $dx = -2 \sin(\theta) d\theta$. Thus

$$\mu_{\text{ST}} = \mathbb{1}_{[0, \pi]} \frac{1}{\pi} \sqrt{1 - \cos^2(\theta)} (-2 \sin(\theta)) d\theta = \mathbb{1}_{[0, \pi]} \left(-\frac{2}{\pi} \sin^2(\theta) \right) d\theta. \quad (232)$$

Thus

$$\begin{aligned} \langle U_m, U_n \rangle &= \int_0^\pi \frac{\sin((m+1)\theta)}{\sin(\theta)} \frac{\sin((n+1)\theta)}{\sin(\theta)} \left(\frac{2}{\pi} \sin^2(\theta) \right) d\theta \\ &= \frac{2}{\pi} \int_0^\pi \sin((m+1)\theta) \sin((n+1)\theta) d\theta \\ &= \begin{cases} 1 & \text{if } m = n \\ 0 & \text{otherwise} \end{cases} \end{aligned} \quad (233)$$

□

Remark 2.4.4 In fact, by Peter-Weyl theory shows that for any compact group K , the character $\text{tr}(\rho)$ of a irreducible unitary representation $\rho : K \rightarrow \text{GL}_r(\mathbb{C})$ form a orthonormal basis of the space $L^2(\text{conjugacy classes of } K, \text{haar measure of } K)$. If $K = \text{SU}_2(\mathbb{C})$, then $\tilde{X} = \{\text{conjugacy classes of } X\} = \left\{ \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}; \theta \in [0, \pi] \right\}$ and then the Haar measure is $\frac{1}{\pi} \sin^2(\theta) d\theta$ and $\tilde{X} \xrightarrow{\text{tr}} [-2, 2]$ becomes a homeomorphism $\theta \rightarrow 2 \cos(\theta)$ and the Haar measure becomes μ_{ST} . Now you can compute the irreducible representations of $\text{SU}_2(\mathbb{C})$; they are parametrized by $n \geq 0$ with character $U_n(2 \cos(\theta))$

Lemma 2.4.5 For $f \in \mathcal{H}_k$ and $\forall n, p$

$$U_n(\lambda_f(p)) = \lambda_f(p^n) \quad (234)$$

Proof Note

$$\sum_{n=0}^{\infty} \lambda_f(p^n) x^n = \frac{1}{1 - \lambda_f(p)x + x^2}. \quad (235)$$

Then $1 - \lambda_f(p)x + x^2 = (1 - \alpha_p x)(1 - \beta_p x)$. Write $\alpha_p = e^{i\theta_p}$ and $\beta_p = e^{-i\theta_p}$, thus $\lambda_f(p) = 2 \cos(\theta_p)$ (θ_p may not be in $\mathbb{R}$). Thus

$$\begin{aligned}
\sum_{n=0}^{\infty} \lambda_f(p^n) x^n &= \frac{1}{(1 - \alpha_p x)(1 - \beta_p x)} \\
&= \frac{1}{(1 - e^{i\theta_p} x)(1 - e^{-i\theta_p} x)} \\
&= \sum_{n=0}^{\infty} \left(\sum_{j=0}^n \alpha_p^j \beta_p^{n-j} \right) x^n
\end{aligned} \tag{236}$$

Thus

$$\begin{aligned}
\lambda_f(p^n) &= \sum_{j=0}^n \alpha_p^j \beta_p^{n-j} \\
&= \sum_{j=0}^n e^{ij\theta_p} e^{-(n-j)\theta_p} \\
&= \sum_{j=0}^n e^{i(2j-n)\theta_p} \\
&= e^{-in\theta_p} \sum_{j=0}^n e^{2ij\theta_p} \\
&= e^{-in\theta_p} \frac{e^{2i(n+1)\theta_p} - 1}{e^{2i\theta_p} - 1} \\
&= \frac{\sin((n+1)\theta_p)}{\sin(\theta_p)} = U_n(2\cos(\theta_p)) = U_n(\lambda_f(p))
\end{aligned} \tag{237}$$

□

The lemma shows that the study of this is broadly linked to the irreducible representations of $\mathrm{SU}_2(\mathbb{C})$

Theorem 2.4.6 (*Serre, Sarnak, K-Saha-Tsinmann*) Let k be s.t $S_k(1) \neq 0$ so $\mathcal{H}_k \neq \emptyset$. Let p be a prime number. For $f \in \mathcal{H}$, let $\lambda_f(p) = \frac{a_f(p)}{p^{\frac{k-1}{2}}}$. Let

$$\mu_k = \frac{\Gamma(k-1)}{(4\pi)^{k-1}} \sum_{f \in \mathcal{H}_k} \frac{1}{\|f\|^2} \delta_{\lambda_f(p)} \tag{238}$$

this is a measure on $\mathbb{R}$, supported on the compact interval since $|\lambda_f(p)| = O(\sqrt{p})$. Then μ_k weakly converges to **Sato-Tate Measure**

$$\mu_{\mathrm{ST}} = \mathbb{1}_{[-2,2]} \frac{1}{\pi} \sqrt{1 - \frac{x^2}{4}} \, dx \tag{239}$$

Proof Using the Proposition 5.4.1 (Weyl Criterion), we want a set of test functions $\varphi : [-2, 2] \rightarrow \mathbb{C}$ which span a dense subset, whose integral converges to 0 in the limit (its just easier to prove that something converges to 0 than some other limit), we also want the functions to “interact well” with μ_k . Let X be some large interval containing all $\lambda_f(p)$ and $\mu = \mu_{\mathrm{ST}}$. One can take as test functions $\varphi_n(x) = x^n$ for $n \geq 0$ which span a dense subset

of $\mathcal{C}(X)$ (“method of moments”). You can do this but not really optimal. A better choice is $(U_n)_{n \in \mathbb{N}}$ which are “Chebyshev polynomials of the 2nd kind”. Where $U_n(2 \cos \theta) = \frac{\sin((n+1)\theta)}{\sin(\theta)}$ for $\theta \in \mathbb{R}$ or $\mathbb{C}$. You can show that $U_n \in \mathbb{R}[X]$ and $\deg U_n = n$ and $U_0 = 1$. Then U_n span the same subspace of $\mathcal{C}(X)$ as (X^n) , so a dense subspace.

$$\begin{aligned} \int_{-2}^2 U_n(x) d\mu_k &= \frac{\Gamma(k-1)}{(4\pi)^{k-1}} \sum_{f \in \mathcal{H}_k} \frac{1}{\|f\|^2} U_n(\lambda_f(p)) \\ &= \frac{\Gamma(k-1)}{(4\pi)^{k-1}} \sum_{f \in \mathcal{H}_k} \frac{1}{\|f\|^2} \lambda_f(p^n) \end{aligned} \quad (240)$$

(A quick remark, if the the Sarnak Serre measure (see below) was used then we would get $\frac{1}{|H_k|} \frac{1}{p^{\frac{k-1}{2}}} \text{tr}(T(p^n))$ which is why the selberg trace formula would be needed.) Now notice that $\tilde{H}_k = \left\{ \frac{f}{\|f\|} : f \in \mathcal{H}_k \right\}$ is a orthonormal basis of $S_k(1)$. So we have

$$\begin{aligned} \int_{-2}^2 U_n(x) d\mu_k &= \frac{\Gamma(k-1)}{(4\pi)^{k-1}} \sum_{f \in \mathcal{H}_k} \frac{a_f(p^n)}{p^{\frac{n(k-1)}{2}}} \\ &= \frac{\Gamma(k-1)}{(4\pi)^{k-1} p^{\frac{n(k-1)}{2}}} \sum_{f \in \tilde{H}_k} a_f(p^n) \\ &= \frac{\Gamma(k-1)}{(4\pi)^{k-1} p^{\frac{n(k-1)}{2}}} \sum_{f \in \tilde{H}_k} a_f(p^n) a_f(1) \quad (\text{Because } a_f(1) = 1) \\ &= \delta(p^n, 1) + \frac{2\pi}{i^k} \sum_{c \geq 1} \frac{1}{c} S(p^n, 1; c) J_{k-1} \left(\frac{4\pi p^{\frac{n}{2}}}{c} \right) \quad (\text{Because of Petersson Formula}) \\ &= \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases} + \text{remainder} \end{aligned} \quad (241)$$

we want to show that as $k \rightarrow \infty$, the remainder goes to 0. Good thing for us is that nothing in the remainder depends on k other than the bessel function, so all we need to do is to show that $k \rightarrow \infty \Rightarrow J_{k-1} \rightarrow 0$ for $x \geq 0$. If we look back to the proof of Petersson Formula we have the fact that $\int U_n \mu_k = p^n$ th Fourier coefficient of $P_{1,k}$. We have

$$a_m(P_{m,k}) = \int_0^1 P_{m,k}(x+i) e(-m(x+i)) dx \quad (242)$$

by definition. Recall definition of $P_{m,k}$,

$$P_{m,k}(z) = \sum_{g \in \overline{T}/\text{SL}_2(\mathbb{Z})} j_k(g, z) e(ngz) \quad (243)$$

Note that $|j_k(g, z) e(ngz)| \leq |j_k(g, z)|$ and $|cz + d|^2 = |c(x+i) + d| \geq c^2$. So for $c \neq 0$ we get

$$|j_k(g, z) e(ngz)| \xrightarrow[k \rightarrow \infty]{} 0 \quad (244)$$

uniformly for $g \in \overline{T}/\mathrm{SL}_2(\mathbb{Z})$, $c \neq 0$ and $z = x + i$. For $c = 0$, we have $g = I(\text{class mod } \overline{T})$ so $e(ngz)j(g, z) = e(nz)$. Since moreover $|e(ngz)j_k(g, z)| \leq j_4(g, z)$ for $k \geq 4$, which defines a convergent series it follows by dominated convergence that $P_{n,k} \xrightarrow{k \rightarrow \infty} e(nz)$.

$$\begin{aligned} a_m(P_{m,k}) &= \int_0^1 P_{m,k}(x+i)e(-m(x+i)) \, dx \xrightarrow{k \rightarrow \infty} \int_0^1 e((n-m)(x+i)) \, dx \\ &= \delta(m, n) \end{aligned} \quad (245)$$

Thus the remainder of $\int U_n(x) \, d\mu_k \rightarrow 0$ as $k \rightarrow \infty$ □

Corollary 2.4.7 The set $\{\lambda_f(p) \mid f \in \mathcal{H}_k \text{ for some } k\}$ is dense in $[-2, 2]$ (closure contains in $[-2, 2]$)

Remark 2.4.8

1. This shows that that $|\lambda_f(p)| \leq 2$ “most of the time”. Deligne proved this was always true in a theorem found in 1974 where he proved the Ramanujan Petersson conjecture. But this proof is much harder and requires details from the finite field Riemann Hypothesis.
2. Natural question to ask is, what about $\tilde{\mu}_k = \frac{1}{|\mathcal{H}_k|} \sum_{f \in \mathcal{H}_k} \delta_{\lambda_f(p)}$? Sarnak and Serre worked on this and proved that

$$\tilde{\mu}_k \xrightarrow{k \rightarrow \infty} \frac{p+1}{\pi} \frac{\sqrt{1 + \frac{x^2}{4}}}{\left(\sqrt{p} + \frac{1}{\sqrt{p}}\right)^2 - x^2} \mathbb{1}_{[-2,2]} \, dx \quad (246)$$

which is the “Plancherel measure for $\mathrm{SL}_2(\mathbb{Q}_p)$ ”. To prove this they used the Selberg Trace formula for which gives a “explicit” formula for the trace $\mathrm{tr}(T(n))$ of the Hecke operators.

3. One can also ask what happens if we fix f (eg $f = \Delta \in \mathrm{SL}_2(\mathbb{Z})$) and vary p ? This is the problem which was the Sato Tate conjecture. For $f \in \mathcal{H}_k$ fixed and for all functions $\varphi : [-2, 2] \rightarrow \mathbb{C}$ continuous,

$$\frac{1}{\pi} \sum_{p \leq x} \varphi(\lambda_f(p)) \rightarrow \int_{-2}^2 d\mu_{\mathrm{ST}} \quad (247)$$

3. Theta Functions

Theta functions are functions on $\mathbb{H}$ which give rise to Automorphic forms which are quite different from those already constructed. They are related to integral positive-definite quadratic forms and their modularity properties for representations of integers by such quadratic forms like sums of squares.

Definition 3.1 $Q : \mathbb{R}^n \rightarrow \mathbb{R}$ is a **positive definite quadratic form** if $\exists A \in M_r(\mathbb{Z})$ where A is symmetric where

$$Q(x) = x^t A x = \sum_i^n a_{ii} x_i^2 + 2 \sum_{1 \leq i < j \leq n} a_{i,j} x_i x_j. \quad (248)$$

Thus if $x \neq 0 \Rightarrow Q(x) > 0$. Q is **even** if $\forall i \in [1, \dots, n], a_{ii} \equiv 0 \pmod{2}$, in which case $Q(x) \equiv 0 \pmod{2}$ for all $x \in \mathbb{Z}^n$.

Definition 3.2 The “basic” theta function $\theta_Q : \mathbb{H} \rightarrow \mathbb{C}$ associated with a positive definite quadratic form Q is

$$\theta_Q(z) = \sum_{m \in \mathbb{Z}^n} e\left(\frac{Q(m)z}{2}\right). \quad (249)$$

A more general quadratic form is the following

$$\theta_Q(z; f) = \sum_{m \in \mathbb{Z}^n} e\left(\frac{Q(m)z}{2}\right) f(m) \quad (250)$$

for some $f : \mathbb{Z}^n \rightarrow \mathbb{C}$

Example 3.3 Let $Q(x) = x^2$, which is the simplest quadratic form.

$$\theta_Q(z) = \sum_{m \in \mathbb{Z}} e\left(\frac{m^2 z}{2}\right) = \sum_{m \in \mathbb{Z}} e^{im^2 \pi z} = 1 + 2 \sum_{m=1}^{\infty} e^{im^2 \pi z} \quad (251)$$

3.1. “One variable” theta functions and Dirichlet L function

We will look at the following theta function for $Q(x) = x^2$:

$$\theta_Q(z; \chi) = \sum_{m \in \mathbb{Z}} \chi(m) e\left(\frac{m^2 z}{2}\right) \quad (252)$$

for some Dirichlet character $\chi : \mathbb{Z}_q^* \rightarrow \mathbb{C}$. and

$$\theta_Q(z; a) = \sum_{m \in \mathbb{Z}} \mathbb{1}_{m \equiv a \pmod{q}} e\left(\frac{m^2 z}{2}\right) = \sum_{m \in a + q\mathbb{Z}} e\left(\frac{m^2 z}{2}\right) \quad (253)$$

Lemma 3.1.1 $\theta_Q(z; \chi)$ and $\theta_Q(z; a)$ are well defined holomorphic functions on $\mathbb{H}$ such that

$$\begin{aligned} \theta_Q(z + 2; \chi) &= \theta_Q(z; \chi) \\ \theta_Q(z + 2; a) &= \theta_Q(z; a) \end{aligned} \quad (254)$$

Proof Note for $z = x + iy \in \mathbb{H}$.

$$\left| e\left(\frac{m^2 z}{2}\right) \right| = \left| e^{im^2 \pi(x+iy)} \right| = e^{-im^2 y} \quad (255)$$

thus the series $\sum_{m \in \mathbb{Z}} e\left(\frac{m^2 z}{2}\right)$ converges locally absolutely, thus its a holomorphic function on $\mathbb{H}$. The 2 periodicity is obvious:

$$e\left(\frac{m^2(z+2)}{2}\right) = e\left(\frac{m^2 z}{2} + m^2\right) = e\left(\frac{m^2 z}{2}\right) \quad (256)$$

□

Lemma 3.1.2 (Jacobi Transformation Part 1) Let $y > 0$ be a real number. We have

$$\theta_Q(iy; a) = \frac{1}{q\sqrt{y}} \sum_{x=0}^{q-1} e\left(\frac{ax}{q}\right) \theta_Q\left(-\frac{1}{q^2 iy}; x\right) \quad (257)$$

Proof By definition,

$$\begin{aligned} \theta_Q(iy; a) &= \sum_{m \in a + \mathbb{Z}q} e^{-m^2 \pi y} \\ &= \sum_{n \in \mathbb{Z}} e^{-(a+nq)^2 \pi y} \end{aligned} \quad (258)$$

Let $f_y : \mathbb{R} \rightarrow \mathbb{C}$ where $n \rightarrow e^{-(a+nq)^2 \pi y}$. f_y has rapid decay and is smooth, thus the poisson summation formula could be used.

Thus

$$\begin{aligned} \theta_Q(iy; a) &= \sum_{n \in \mathbb{Z}} f_y(n) \\ &= \sum_{h \in \mathbb{Z}} \hat{f}(h) \end{aligned} \quad (259)$$

$$\begin{aligned}
\hat{f}(t) &:= \int_{-\infty}^{\infty} f(x) e^{-xt} dt \\
&= \int_{-\infty}^{\infty} e^{-\pi(a+xq)^2 y} e^{-xt} dx \\
&= \frac{1}{q} \int_{-\infty}^{\infty} e^{-\pi u^2 y} e^{\frac{(a-u)t}{q}} du \quad (u = a + xq) \\
&= \frac{1}{q} e^{\frac{at}{q}} \int_{-\infty}^{\infty} e^{-\pi u^2 y} e^{-\frac{ut}{q}} du \\
&= \frac{e^{\frac{at}{q}}}{q\sqrt{y}} \int_{\mathbb{R}} e^{-\pi v^2} e^{-\frac{vt}{q\sqrt{y}}} dv \quad (v^2 = u^2 y) \\
&= \frac{e^{\frac{at}{q}}}{q\sqrt{y}} \hat{g}\left(\frac{t}{q\sqrt{y}}\right)
\end{aligned} \tag{260}$$

where $g(x) = e^{-\pi x^2}$. The fourier transform of a gaussian is a gaussian and in this case it stays fixed thus $\hat{g} = g$. Thus

$$\hat{f}(t) = \frac{e^{\frac{at}{q}}}{q\sqrt{y}} g\left(\frac{t}{q\sqrt{y}}\right). \tag{261}$$

Plugging the result back into θ_Q we see

$$\begin{aligned}
\theta_Q(z; a) &= \frac{1}{q\sqrt{y}} \sum_{h \in \mathbb{Z}} e^{\frac{ah}{q}} g\left(\frac{h}{q\sqrt{y}}\right) \\
&= \frac{1}{q\sqrt{y}} \sum_{h \in \mathbb{Z}} e^{\frac{ah}{q}} e^{-\frac{\pi h^2}{q^2 y}} \\
&= \frac{1}{q\sqrt{y}} \sum_{h \in \mathbb{Z}} e^{\frac{ah}{q}} e^{-\frac{2\pi i h^2}{2q^2 i y}} \\
&= \frac{1}{q\sqrt{y}} \sum_{h \in \mathbb{Z}} e^{\frac{ah}{q}} e\left(-\frac{h^2}{2q^2(iy)}\right) \\
&= \frac{1}{q\sqrt{y}} \sum_{x=0}^{q-1} \sum_{h \in \mathbb{Z}} e^{\frac{a(x+hq)}{q}} e\left(-\frac{(x+hq)^2}{2q^2(iy)}\right) \\
&= \frac{1}{q\sqrt{y}} \sum_{x=0}^{q-1} e^{\frac{ax}{q}} \sum_{h \in x+q\mathbb{Z}} e\left(-\frac{h^2}{2q^2(iy)}\right) \\
&= \frac{1}{q\sqrt{y}} \sum_{x=0}^{q-1} e^{\frac{ax}{q}} \theta_Q\left(-\frac{1}{iyq^2}; x\right)
\end{aligned} \tag{262}$$

□

Lemma 3.1.3 (*Jacobi Transformation Part 2*) Let $\chi : \mathbb{Z} \rightarrow \mathbb{C}$ is a primitive dirichlet character mod q .

$$\theta_Q(iy; \chi) = \frac{\tau(\chi)}{q\sqrt{y}} \theta_Q\left(-\frac{1}{q^2 iy}; \bar{\chi}\right) \quad (263)$$

where

$$\tau(\chi) := \sum_{x=0}^{q-1} \chi(x) e\left(\frac{x}{q}\right) \quad (264)$$

Proof

$$\begin{aligned} \theta_Q(z; \chi) &= \sum_{m \in \mathbb{Z}} \chi(m) e\left(\frac{m^2 z}{2}\right) \\ &= \sum_{a=0}^{q-1} \chi(a) \sum_{m \in a+q\mathbb{Z}} e\left(\frac{m^2 z}{2}\right) \\ &= \sum_{a=0}^{q-1} \chi(a) \theta_Q(z; a) \end{aligned} \quad (265)$$

Thus by Lemma 3.1.2,

$$\begin{aligned} \theta_Q(iy; \chi) &= \sum_{a=0}^{q-1} \chi(a) \theta_Q(iy; a) \\ &= \frac{1}{q\sqrt{y}} \sum_{a=0}^{q-1} \chi(a) \sum_{b=0}^{q-1} e^{\frac{ab}{q}} \theta_Q\left(-\frac{1}{iyq^2}; b\right) \\ &= \frac{1}{q\sqrt{y}} \sum_{b=0}^{q-1} \theta_Q\left(-\frac{1}{iyq^2}; b\right) \sum_{a=0}^{q-1} \chi(a) e^{\frac{ab}{q}} \end{aligned} \quad (266)$$

Now a property of primitive characters is that $\sum_{a=0}^{q-1} \chi(a) e^{\frac{ab}{q}} = \tau(\chi) \overline{\chi(b)}$.

Thus

$$\begin{aligned} \theta_Q(iy; \chi) &= \frac{\tau(\chi)}{q\sqrt{y}} \sum_{x=0}^{q-1} \overline{\chi(x)} \theta_Q\left(-\frac{1}{iyq^2}; x\right) \\ &= \frac{\tau(\chi)}{q\sqrt{y}} \sum_{x=0}^{q-1} \overline{\chi(x)} \sum_{m \in x+q\mathbb{Z}} e^{-\pi i m^2 \left(\frac{1}{yq^2}\right)} \\ &= \frac{\tau(\chi)}{q\sqrt{y}} \sum_{m \in \mathbb{Z}} \bar{\chi}(m) e^{-\pi i m^2 \left(\frac{1}{yq^2}\right)} = \frac{\tau(\chi)}{q\sqrt{y}} \theta_Q\left(-\frac{1}{iyq^2}; \bar{\chi}\right) \end{aligned} \quad (267)$$

□

Corollary 3.1.4 (*L-Function of Primitive Character and Lambda Functional*) Let $q \geq 1$ be a integer and χ a dirichlet character mod q which is primitive.

1. If $q = 1$, $L(s, \chi) = \sum_{n \geq 1} \frac{1}{n^s} \chi(n) = \zeta(s)$, then $\zeta(s)$ has a analytic continuation to a meromorphic function on $\mathbb{C}$, with a unique simple pole at $s = 1$ with residue 1, and which satisfies

$$\pi^{-\frac{s}{2}} \zeta\left(\frac{s}{2}\right) \zeta(s) = \pi^{-\frac{1-s}{2}} \zeta\left(\frac{1-s}{2}\right) \zeta(1-s) \quad (268)$$

2. If $q \geq 2$, then $L(s, \chi)$ has a analytic continuation to an **entire** function, and moreover the function

$$\Lambda(\chi, s) = \pi^{-\frac{s+t_\chi}{2}} \Gamma\left(\frac{s+t_\chi}{2}\right) L(\chi, s) \quad (269)$$

satisfies the functional equation

$$\Lambda(\chi, s) = \frac{1}{i^{t_\chi}} \frac{\tau(\chi)}{\sqrt{q}} q^{\frac{1}{2}-s} \Lambda(\bar{\chi}, 1-s) \quad (270)$$

where

$$t_\chi = \begin{cases} 0 & \text{if } \chi(-1) = 1 \\ 1 & \text{if } \chi(-1) = -1 \end{cases} \quad (271)$$

We will prove this for $\chi(-1) = 1$.

Proof

$$\begin{aligned} \Lambda(\chi, s) &= \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \sum_{n=1}^{\infty} n^{-s} \chi(n) \\ &= \sum_{n=1}^{\infty} \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) n^{-s} \chi(n) \\ &= \sum_{n=1}^{\infty} \int_0^{\infty} \chi(n) e^{-\pi n^2 x} x^{\frac{s}{2}-1} dx \end{aligned} \quad (272)$$

Last equality comes from Proposition 5.3.2. Integral and summation can be interchanged because of quick convergence.

$$\begin{aligned} \Lambda(\chi, s) &= \int_0^{\infty} x^{\frac{s}{2}-1} \sum_{n=1}^{\infty} \chi(n) e^{-\pi n^2 x} dx \\ &= \frac{1}{2} \int_0^{\infty} x^{\frac{s}{2}-1} \theta_Q(ix; \chi) dx \end{aligned} \quad (273)$$

This is because $\chi(-n) = \chi(-1)\chi(n) = \chi(n)$ since χ is even.

We will now use the jacobi transformation to solve the integral much in the same way we solved the similar integral for regular modular forms.

$$\begin{aligned}
\Lambda(\chi, s) &= \frac{1}{2} \int_0^\infty x^{\frac{s}{2}-1} \theta_Q(ix; \chi) dx \\
&= \frac{1}{2} \left(\int_0^{\frac{1}{q}} x^{\frac{s}{2}-1} \theta_Q(ix; \chi) dx + \int_{\frac{1}{q}}^\infty x^{\frac{s}{2}-1} \theta_Q(ix; \chi) dx \right)
\end{aligned} \tag{274}$$

Let's deal with the first integral. By the Jacobi transformation Lemma 3.1.3,

$$\int_0^{\frac{1}{q}} x^{\frac{s}{2}-1} \theta_Q(ix; \chi) dx = \frac{\tau(\chi)}{q} \int_0^{\frac{1}{q}} x^{\frac{s-1}{2}} \theta_Q\left(\frac{i}{xq^2}; \bar{\chi}\right) \frac{dx}{x} \tag{275}$$

Let $u = \frac{1}{q^2 x}$ and $du = -\frac{1}{q^2 x^2} dx \Rightarrow \frac{du}{u} = q^2 x du = -\frac{dx}{x}$. Thus

$$\begin{aligned}
\int_0^{\frac{1}{q}} x^{\frac{s}{2}-1} \theta_Q(ix; \chi) dx &= \frac{\tau(\chi)}{q} \int_{\frac{1}{q}}^\infty \left(\frac{1}{q^2 u} \right)^{\frac{s-1}{2}} \theta_Q(iu; \bar{\chi}) \frac{du}{u} \\
&= \tau(\chi) q^{-s} \int_{\frac{1}{q}}^\infty u^{-\frac{s-1}{2}} \theta_Q(iu; \bar{\chi}) du
\end{aligned} \tag{276}$$

Thus

$$\Lambda(\chi, s) = \frac{1}{2} \int_{\frac{1}{q}}^\infty \left(\frac{\tau(\chi)}{\sqrt{q}} q^{\frac{1}{2}-s} x^{\frac{1-s}{2}} \theta_Q(ix; \bar{\chi}) + x^{\frac{s}{2}} \theta_Q(ix; \chi) \right) dx \tag{277}$$

This gives analytic continuation and the functional equation after checking that

$$\tau(\chi)\tau(\bar{\chi}) = q\chi(-1) = q \tag{278}$$

For $q = 1$, namely $\zeta(s)$, argument is the similar

$$\begin{aligned}
\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s) &= \frac{1}{2} \int_0^\infty \left(\sum_{n \in \mathbb{Z}/\{0\}} e^{-\pi n^2 x} \right) x^{\frac{s}{2}-1} dx \\
&= \frac{1}{2} \int_0^\infty (\theta_Q(ix) - 1) x^{\frac{s}{2}-1} dx
\end{aligned} \tag{279}$$

After we recombine keeping sure not to integrate $x^{\frac{s}{2}-1}$ close to ∞ the result follows. $\square$

3.2. The Basic Transformation Formula in General

Proposition 3.2.1 Let $Q(x) = x^T A x$ be a quadratic form; let $N \geq 1$ such that NA^{-1} has integral entries. For $z \in \mathbb{H}$, $x \in \mathbb{C}^n$ we have

$$\sum_{m \in \mathbb{Z}^n} e\left(\frac{1}{2}Q(m+x)z\right) = \frac{1}{\det(A)^{\frac{1}{2}}} \left(\frac{i}{z}\right)^{\frac{n}{2}} \sum_{m \in \mathbb{Z}^n} e\left(-\frac{1}{2z}\tilde{Q}(m) + m \cdot x\right) \tag{280}$$

where $\tilde{Q}(x) = x^T A^{-1} x$

Proof $\forall z \in \mathbb{H}$, let $\varphi(y) = e(\frac{1}{2}Q(y)z)$. There is a more general version of the Poisson Summation formula which we will use because of sufficient decay properties of φ .

$$\sum_{m \in \mathbb{Z}^n} \varphi(m+x) = \sum_{h \in \mathbb{Z}^n} \hat{\varphi}(h) e(h \cdot x) \quad (281)$$

Let us calculate the Fourier Coefficients. We want to show that:

$$\hat{\varphi}(t) = \int_{\mathbb{R}^n} \varphi(y) e(-y \cdot t) dy = \frac{1}{\det(A)^{\frac{1}{2}}} \left(\frac{i}{z} \right)^{\frac{n}{2}} e\left(-\frac{1}{2z} \tilde{Q}(t)\right) \quad (282)$$

By analytic continuation, it suffices to look at the integrals for $z = iw \in \mathbb{H}$. By linear algebra, exists a basis of $\mathbb{R}^n$ such that $u : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a invertible linear map and $Q(y) = |u(y)|^2 = \sum_{i=1}^n u_i(y)^2$ for $y \in \mathbb{R}^n$. Thus

$$\begin{aligned} \hat{\varphi}(t) &= \int_{\mathbb{R}^n} e\left(\frac{1}{2}Q(y)z\right) e(-y \cdot t) dy \\ &= \int_{\mathbb{R}^n} e\left(\frac{1}{2}|u(y)|^2 z\right) e(-y \cdot t) dy \end{aligned} \quad (283)$$

Let $v = u(y)$ and $dv = |\det(u)| dy$.

$$\begin{aligned} \hat{\varphi}(t) &= \frac{1}{|\det(u)|} \int_{\mathbb{R}^n} e\left(\frac{1}{2}|v|^2 z\right) e(-u^{-1}(v) \cdot t) dv \\ &= \frac{1}{|\det(u)|} \int_{\mathbb{R}^n} e\left(\frac{1}{2}|v|^2 z - u^{-1}(v) \cdot t\right) dv \\ &= \frac{1}{|\det(u)|} \int_{\mathbb{R}^n} e\left(\frac{1}{2}|v|^2 z - v \cdot (u^{-1})^\dagger t\right) dv \\ &= \frac{1}{|\det(u)|} \prod_{j=1}^n \frac{1}{\sqrt{w}} e\left(-\frac{1}{2iw} \left((u^{-1})^\dagger t\right)^2\right) \end{aligned} \quad (284)$$

The last statement comes from Fubini's Theorem.

$$\begin{aligned} \hat{\varphi}(t) &= \frac{1}{|\det(u)|} \prod_{j=1}^n \frac{i}{\sqrt{w}} e\left(-\frac{1}{2iw} \left((u^{-1})^\dagger t\right)^2\right) \\ &= \frac{1}{|\det(u)|} \left(\frac{i}{z}\right)^{\frac{n}{2}} e\left(-\frac{1}{2iw} \left|(u^{-1})^\dagger t\right|^2\right) \end{aligned} \quad (285)$$

Let B be the matrix such that $u(y) = By$. Then $B^\dagger B = A$ (linear algebra). Then

$$\left|(u^{-1})^\dagger t\right|^2 = \tilde{Q}(t) \quad (286)$$

When you plug everything back in you get the statement of the proposition. $\square$

Definition 3.2.2 A **Q-spherical polynomial** is a polynomial $P \in \mathbb{C}[x_1, \dots, x_n]$ which is a linear combination of polynomials of the form

1. $P = \text{constant}$
2. $P = \text{Linear form}$
3. $P(x) = (c^T Ax)^\nu$ for $\nu \geq 2$ and $c \in \mathbb{C}^n \ni Q(c) = 0$

Proposition 3.2.3 Let $Q(x) = x^T A x$ be a positive definite quadratic form and P a Q -spherical polynomial. Then $\tilde{P}(x) = P(A^{-1}x)$ is a $\tilde{Q}$ -spherical polynomial.

$$\sum_{m \in \mathbb{Z}^n} P(m+x) e\left(\frac{z}{2} Q(m+x)\right) = \frac{1}{\deg(P)} \frac{1}{\sqrt{\det(A)}} \left(\frac{i}{z}\right)^{\frac{n}{2} + \deg(P)} \sum_{h \in \mathbb{Z}^n} \tilde{P}(h) e\left(-\frac{Q(h)}{2z} + (287)x\right)$$

Theorem 3.2.4 Let $Q(x) = |x|^2 = x^T I x$. Let P be a Q spherical polynomial homogeneous of degree ν . Let $\theta = \theta_Q(\cdot, P) \in M_{\nu + \frac{n}{2}}(\Gamma(4))$ and if P is nonconstant $\theta \in S_{\nu + \frac{n}{2}}(\Gamma(4))$

Proof Note $\Gamma(4) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z}) \mid \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv I \pmod{4} \right\}$. and $\tilde{P} = P$. Let $k = \frac{n}{2} + \nu$

We will first find the value of $\theta\left(-\frac{1}{z}\right)$.

$$\begin{aligned} \theta\left(-\frac{1}{z}\right) &= \sum_{m \in \mathbb{Z}^n} P(m) e\left(-\frac{1}{2z} Q(m)\right) \\ &= \frac{1}{\nu} \frac{1}{\sqrt{\det(I)}} (-iz)^k \sum_{h \in \mathbb{Z}^n} \tilde{P}(h) e\left(\frac{Q(h)z}{2}\right) \\ &= \frac{1}{\nu} (-iz)^k \theta(z) \end{aligned} \tag{288}$$

$\forall \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(4)$. Let $\tilde{\gamma} = \gamma \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} b & -a \\ d & -c \end{pmatrix}$. Then $\tilde{\gamma}z = \frac{bz-a}{dz-c} = \frac{b}{d} - \frac{1}{d(dz-c)}$. Thus

$$\begin{aligned} \theta(\tilde{\gamma}z) &= \sum_{m \in \mathbb{Z}^n} P(m) e\left(\frac{|m|^2}{2} \left(\frac{b}{d} - \frac{1}{d(dz-c)}\right)\right) \\ &= \sum_{m \in \mathbb{Z}^n} P(m) e\left(\frac{|m|^2 b}{2d}\right) e\left(-\frac{|m|}{2d(dz-c)}\right) \end{aligned} \tag{289}$$

The value of $e\left(\frac{|m|^2 b}{2d}\right)$ only depends on $m \pmod{2d}$, and even modulo d since b is even. So

$$\begin{aligned} \theta(\tilde{\gamma}z) &= \sum_{g \in \mathbb{Z}_d^n} e\left(\frac{|g|^2 b}{2d}\right) \sum_{m \in g + d\mathbb{Z}^n} P(m) e\left(-\frac{|m|}{2d(dz-c)}\right) \\ &= \sum_{g \in \mathbb{Z}_d^n} e\left(\frac{|g|^2 b}{2d}\right) \sum_{m \in g + d\mathbb{Z}^n} P(m) e\left(-\frac{d|m|}{2d^2(dz-c)}\right) \end{aligned} \tag{290}$$

□

4. Ranklin Selberg

Earlier we found the bound $|a_f(n)| \ll n^{\frac{k}{2}}$ for $f \in S_k(q, \chi)$. We will improve this estimate by tightening the estimate

$$\sum_{n \leq X} |a_f(n)|^n \ll X^k \quad (291)$$

from which we deduced it. To do this, we will use the basic method of averages of arithmetic functions based on generating Dirichlet series. We will study

$$D(s) \sum_{n=1}^{\infty} \frac{|a_n|^2}{n^s} \quad (292)$$

and its analytic properties. Note that, we at least have absolute locally uniform convergence as soon as $\Re(s) > k + 1$. But we need analytic continuation beyond this region to do better.

Theorem 4.1 For $\Re(s)$ large enough, we have

$$D(s + k - 1) = \frac{(4\pi)^{s+k-1}}{\Gamma(s + k - 1)} \int_{D_F} y^k |f(x + iy)|^2 E(z, s) \frac{dx dy}{y^2} \quad (293)$$

where $E(z, s)$ is a nonholomorphic Eisenstein series defined by

$$E(z, s) = \sum_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} \mathfrak{I}(gz)^s \quad (294)$$

which converges absolutely for $\Re(s) > 2$

Proof Recall

$$\mathfrak{I}(gz) = |j_1(g, z)|^{-2} \mathfrak{I}(z) \quad (295)$$

so in terms of modulus,

$$|\mathfrak{I}(gz)|^s = \left(|j_1(g, z)|^{-2} \mathfrak{I}(z) \right)^{\Re(s)} \quad (296)$$

which defines a absolutely locally uniformly convergent series for $2\Re(s) > 2$, much like the convergence of the Eisenstein series. Thus for the integral

$$\begin{aligned} \int_{D_F} y^k |f(x + iy)|^2 E(z, s) \frac{dx dy}{y^2} &= \int_{D_F} y^k |f(x + iy)|^2 \sum_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} \mathfrak{I}(gz)^s \frac{dx dy}{y^2} \\ &= \sum_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} \int_{D_F} y^k |f(x + iy)|^2 \mathfrak{I}(gz)^s \frac{dx dy}{y^2} \end{aligned} \quad (297)$$

Since $y^k |f(x + iy)|^2$ and $\frac{dx dy}{y^s}$ is $\mathrm{SL}_2(\mathbb{Z})$ invariant,

$$\begin{aligned}
\int_{D_F} y^k |f(x+iy)|^2 E(z, s) \frac{dx dy}{y^2} &= \sum_{g \in \overline{T} \backslash \mathrm{SL}_2(\mathbb{Z})} \int_{g^{-1} D_F} y^k |f(x+iy)|^2 y^s \frac{dx dy}{y^2} \\
&= \int_{D_{\overline{T}/\mathrm{SL}_2(\mathbb{Z})}} y^{k+s} |f(x+iy)|^2 \frac{dx dy}{y^2} \\
&= \int_{D_{\overline{T}/\mathrm{SL}_2(\mathbb{Z})}} y^{k+s} \left| \sum_{n=1}^{\infty} a_f(n) e(n(x+iy)) \right|^2 \frac{dx dy}{y^2} \\
&= \int_0^1 \int_0^\infty y^{k+s} \left| \sum_{n=1}^{\infty} a_f(n) e(n(x+iy)) \right|^2 \frac{dx dy}{y^2} \\
&= \int_0^1 \int_0^\infty y^{k+s} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_f(n) \overline{a_f(m)} e(n(x+iy)) \overline{e(m(x+iy))} \frac{dx dy}{y^2} \\
&= \int_0^1 \int_0^\infty y^{k+s} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_f(n) \overline{a_f(m)} e(n(x+iy)) e(\overline{m(x+iy)}) \frac{dx dy}{y^2}
\end{aligned} \tag{298}$$

□

5. Appendix

5.1. Poisson Summation Formula

Theorem 5.1.1 (*Poisson Summation Formula*) Let $\varphi : \mathbb{R} \rightarrow \mathbb{C}$ be a integrable, C^1 function such that

$$\sum_{n \in \mathbb{Z}} \varphi(n + z) \quad (299)$$

and

$$\sum_{n \in \mathbb{Z}} \varphi'(n + z) \quad (300)$$

converge locally uniformly for $x \in \mathbb{R}$. Then,

$$\sum_{n \in \mathbb{Z}} \varphi(n) = \sum_{h \in \mathbb{Z}} \hat{\varphi}(h) \quad (301)$$

where

$$\hat{\varphi}(t) = \int_{\mathbb{R}} \varphi(x) e(-xt) \, dt \quad (302)$$

is the **Fourier Transform** of φ .

Proof Let $f : \mathbb{R} \rightarrow \mathbb{C}$ where $z \rightarrow \sum_{n \in \mathbb{Z}} \varphi(z + n)$, so by the assumptions $f \in C^1(\mathbb{R})$; and $f \in L^1(S^1)$, by averaging. Thus,

$$f(x) = \sum_{h \in \mathbb{Z}} \hat{f}(h) e(hx) \quad (303)$$

where

$$\begin{aligned} \hat{f}(h) &= \int_0^1 f(x) e(-hx) \, dx \\ &= \int_0^1 \sum_{n \in \mathbb{Z}} \varphi(x + n) e(-hx) \, dx \\ &= \sum_{n \in \mathbb{Z}} \int_0^1 \varphi(x + n) e(-hx) \, dx && \text{(By Dominated Convergence)} \\ &= \sum_{n \in \mathbb{Z}} \int_n^{n+1} \varphi(u) e(-h(u - n)) \, du && (u = x + n) \quad (304) \\ &= \sum_{n \in \mathbb{Z}} e(hn) \int_n^{n+1} \varphi(u) e(-hu) \, du && (1 \text{ periodicity}) \\ &= \int_{\mathbb{R}} \varphi(u) e(-hu) \, du && \text{(Countable additivity of Lebesgue Measure)} \\ &= \hat{\varphi}(h) \end{aligned}$$

Thus,

$$\sum_{n \in \mathbb{Z}} \varphi(n) = f(0) = \sum_{h \in \mathbb{Z}} \hat{f}(h) = \sum_{h \in \mathbb{Z}} \hat{\varphi}(h) \quad (305)$$

□

5.2. Bruhat Decomposition of $\mathrm{SL}_2(\mathbb{Z})$.

Theorem 5.2.1 (*Bruhat Decomposition of $\mathrm{SL}_2(\mathbb{Z})$*)

$$\mathrm{SL}_2(\mathbb{Z}) = \left(\bigsqcup_{c \geq 1} \left(\bigsqcup_{d \in \mathbb{Z}_c^\times} \overline{T} \begin{pmatrix} * & * \\ c & d \end{pmatrix} T \right) \right) \sqcup \overline{N} \quad (306)$$

Where $\begin{pmatrix} * & * \\ c & d \end{pmatrix}$ is any **fixed** element of $\mathrm{SL}_2(\mathbb{Z})$ with (c, d) as the bottom row.

Proof Let $g \in \mathrm{SL}_2(\mathbb{Z})$, then $g \in \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ where $\alpha\delta - \beta\gamma = 1$.

If $\gamma = 0$, then $g \in \overline{N}$. Suppose $\gamma \neq 0$.

1. $\gamma \geq 1$: Let $n = \frac{\delta - \delta \bmod \gamma}{\gamma}$, thus $\delta = d + \gamma n$, where $d = \delta \bmod \gamma$. Note that since $(\delta, \gamma) = 1 \Rightarrow (d, \gamma) = 1$. Note that

$$\begin{aligned} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} \alpha & \alpha n + \beta \\ \gamma & \gamma n + \delta \end{pmatrix} \\ &= \begin{pmatrix} \alpha & \alpha + \beta n \\ \gamma & d \end{pmatrix} = \begin{pmatrix} \alpha' & \beta' \\ c & d \end{pmatrix} \end{aligned} \quad (307)$$

Let $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be the choice of fixed matrix with bottom row (c, d) in the union above (Equation 306). Since

$$ad - bc = 1 = \alpha'd - \beta'c \Rightarrow d(\alpha' - a) = c(\beta' - b) \quad (308)$$

and $(d, c) = 1$, $d \mid \beta' - b$ and $c \mid \alpha' - a$. Thus $\alpha' = mc + a$ and $\beta' = nd + b$. Thus, $dmc = cnd \Rightarrow m = n$. Thus $\alpha' = mc + a$ and $\beta' = md + b$.

$$\begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \alpha' & \beta' \\ c & d \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \quad (309)$$

Thus,

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & -n \\ 0 & 1 \end{pmatrix} \in T \begin{pmatrix} a & b \\ c & d \end{pmatrix} T \quad (310)$$

2. $\gamma \leq -1$: $-g \in \overline{T} \begin{pmatrix} a & b \\ c & d \end{pmatrix} T \Rightarrow g \in \overline{T} \begin{pmatrix} a & b \\ c & d \end{pmatrix} T$

Since (m, n) were unique, the union is disjoint. □

Corollary 5.2.2 Let $\varphi : \mathbb{H} \rightarrow \mathbb{C}$ be any function such that

$$\sum_{\overline{T}/\mathrm{SL}_2(\mathbb{Z})} (\varphi|_k g)(z) \quad (311)$$

converges absolutely, then

$$\sum_{\bar{T}/\mathrm{SL}_2(\mathbb{Z})} (\varphi|_k g)(z) = \varphi(z) + \sum_{c \geq 1} \sum_{d \in \mathbb{Z}_c^\times} \sum_{m \in \mathbb{Z}} \frac{\varphi\left(\begin{pmatrix} * & * \\ c & d \end{pmatrix} \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} z\right)}{(c(z+m) + d)^k} \quad (312)$$

5.3. Gamma Function Properties

Proposition 5.3.1

$$\int_{\Im(w)=y} w^{-k} e(-w) \, dw = \frac{(2\pi)^k}{i^k} \frac{1}{\Gamma(k)} \quad (313)$$

Proposition 5.3.2

$$\int_0^\infty e^{-\pi n^2 x} x^{\frac{s}{2}-1} \, dx = \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) n^{-s} \quad (314)$$

Proof

$$\begin{aligned} \int_0^\infty e^{-\pi n^2 x} x^{\frac{s}{2}-1} \, dx &= \int_0^\infty e^{-\pi u} \left(\frac{u}{\pi n^2}\right)^{\frac{s}{2}-1} \, du \quad (\text{substitute } u = \pi n^2 x) \\ &= n^{-s} \pi^{-\frac{s}{2}} \int_0^\infty e^{-\pi u} u^{\frac{s}{2}-1} \, du \\ &= n^{-s} \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \end{aligned} \quad (315)$$

□

5.4. Equidistribution

Proposition 5.4.1 (*Weyl Criterion*) If you have $\mu_n \rightarrow \mu$, it is enough to prove that

$$\int_X f \, d\mu = \lim_{n \rightarrow \infty} \int f \, d\mu_n \quad (316)$$

is true for $f : X \rightarrow \mathbb{C}$ which are continuous and span a dense subset of $\mathcal{C}(X)$ which contains 1.

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